

Pediatric Bone Age Prediction

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Abstract

Bone age assessment is a critical diagnostic tool in Pediatric Endocrinology for evaluating growth and identifying developmental disorders. However, the conventional method relies on the subjective interpretation of hand radiographs by specialists, a process that is not only time-consuming but also susceptible to interobserver variability (Berst et al., 2001). To address these limitations, this study presents an automated deep learning system for pediatric bone age prediction.

The project implements an end-to-end pipeline encompassing image preprocessing, model development, and clinical interface deployment. Utilizing the RSNA dataset (Halabi et al., 2019), hand radiographs were normalized and augmented to enhance model robustness (Ornek and Ceylan, 2019). A convolutional neural network (CNN), developed in PyTorch, was trained to perform direct age regression, building upon established architectures for medical image analysis (Simonyan and Zisserman, 2015). To facilitate clinical adoption, an intuitive graphical user interface (GUI) was created using Streamlit, allowing for real-time prediction and interaction.

The developed system was rigorously evaluated using standard performance metrics, with optimization techniques applied to ensure both computational efficiency and scalability (Howard et al., 2017). The resulting system demonstrates a functional proof-of-concept for automated bone age assessment. The modular architecture ensures flexibility and provides a transparent, scalable framework for future development and validation, contributing a foundation for objective assistive tools in clinical workflows (Spampinato et al., 2017).

Keywords: Bone Age Assessment, Deep Learning, Pediatric Radiographs, Convolutional Neural Network, Streamlit GUI

Declaration

No part of this project has been submitted in support of an application for any other degree or qualification at this or any other institute of learning. Apart from those parts of the project containing citations to the work of others, this project is my own unaided work. This work has been carried out in accordance with the Manchester Metropolitan University research ethics procedures and has received ethical approval number 80810.

Signed: Ruchi Rathod

Date: 12/09/25

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Abbreviations

CNN: Convolutional Neural Network

RSNA: Radiological Society of North America

MAE: Mean Absolute Error

RMSE: Root Mean Squared Error

CLAHE: Contrast Limited Adaptive Histogram Equalization

ReLU: Rectified Linear Unit

MSE: Mean Squared Error

TW: Tanner Whitehouse

GP: Greulich and Pyle

FDA: U.S. Food and Drug Administration

Chapter 1: Introduction

Advances in Artificial Intelligence and modern computing have transformed how healthcare challenges are approached, offering opportunities to automate tasks that previously relied solely on clinical expertise (Alzubaidi et al., 2021). In medical imaging, machine learning and deep learning models are increasingly used to detect, classify, and predict conditions with speed and consistency that complement human judgment (Burlina et al., 2017). This shift has been driven by the need to improve diagnostic reliability while reducing the workload of healthcare professionals in time-sensitive clinical environments.

One area where these demands are particularly significant is pediatric bone age assessment. Bone age estimation provides a measure of skeletal maturity and plays a central role in diagnosing growth and endocrine disorders, monitoring abnormal growth trajectories, and guiding interventions such as growth hormone therapy or treatment of early and delayed puberty (Martin et al., 2011a). It also has applications in predicting adult height and in certain clinical contexts where chronological age is uncertain (Poznanski et al., 1978). Bone age is most often determined from radiographs of the left hand and wrist, which reflect predictable stages of bone development across childhood and adolescence. The accuracy of this assessment is crucial, as delayed recognition of growth abnormalities can significantly affect treatment outcomes and a child's long-term health (Lee et al., 2017).

Traditionally, clinicians perform bone age estimation by visually comparing radiographs to reference standards. Two of the most widely used methods are the Greulich & Pyle (GP) atlas (White, 1963), which involves matching the patient's radiograph to a series of standard reference images, and the Tanner–Whitehouse (TW) scoring system (Garn and Tanner, 1986), which evaluates specific bones individually to derive a maturity score. While these approaches remain widely used, they are labour-intensive, dependent on expert knowledge, and prone to significant subjectivity. Numerous studies have shown variability both between observers and for the same observer at different times, leading to inconsistencies in diagnosis and management (Berst et al., 2001). These limitations pose challenges in clinical settings with high patient volumes, where fast, reliable, and reproducible assessments are required.

The development of automated techniques has been motivated by these challenges. Early computational attempts often relied on handcrafted feature extraction, focusing on anatomical

landmarks in radiographs such as epiphyseal plates or bone contours (Bagnall and Davis, 2014). While these approaches offered some degree of standardization, they required domain-specific feature engineering and could not easily adapt to variations in image quality or anatomy. In contrast, deep learning, and specifically convolutional neural networks (CNNs), provides a more powerful alternative by learning relevant features directly from image data (Simonyan and Zisserman, 2015). CNNs have been shown to excel in medical image classification, segmentation, and regression tasks, making them an attractive option for bone age prediction (Iglovikov et al., 2018).

The availability of large, annotated datasets has been a key factor in advancing these techniques. A notable example is the RSNA Pediatric Bone Age Challenge dataset (Halabi et al., 2019), which contains thousands of labeled radiographs used for training and benchmarking automated systems. This resource has enabled the development of models that achieve near-human accuracy and has established a baseline for comparison across studies. By building on such datasets, researchers have moved closer to clinical-grade automated solutions (Spampinato et al., 2017).

This project is situated within this evolving field, aiming to design, develop, and evaluate an end-to-end framework for automated Pediatric bone age prediction. The work focuses on two main components: (1) preprocessing of radiographs using image enhancement and normalisation techniques to ensure consistent input quality, and (2) model development and comparison using multiple CNN-based architectures implemented in PyTorch, to perform direct regression of bone age in months and to evaluate which model performs best. The objective is not only to achieve predictive accuracy comparable to expert radiologists but also to address practical considerations such as efficiency and ease of use, which are essential for integration into clinical workflows. A further contribution of this work is the development of a user-friendly graphical interface that allows non-technical clinical staff to upload X-rays and obtain predictions, demonstrating how such a system could be deployed in practice.

The objectives for this study were as follows:

- To apply preprocessing and augmentation methods that improve the quality and consistency of hand X-ray images for model training.
- To train and compare multiple CNN-based architectures for predicting bone age.

- To evaluate and benchmark the models against the RSNA challenge baseline using clinically relevant metrics such as Mean Absolute Error (MAE) and other performance measures.
- To integrate the best-performing model into a graphical user interface (GUI) to demonstrate usability in a clinical-like setting.
- To assess the performance, strengths, and limitations of the models and discuss directions for future improvement.

The next chapter reviews the clinical background of bone age assessment, traditional manual methods, and prior computational approaches, including recent applications of deep learning in medical imaging. This is followed by Design & Implementation chapters describing dataset preparation, preprocessing strategies, model architecture, and system implementation. The subsequent chapter presents the results of model evaluation and interface testing. A discussion chapter then analyses the findings, comparing them with existing literature, highlighting limitations, and suggesting future research opportunities. The report concludes by summarising the project's contributions and reflecting on its potential implications for Pediatric healthcare.

Chapter 2: Literature Review

2.1 Background: Clinical Relevance of Bone Age and Pediatric Bone Age Assessment

Pediatric Bone Age Assessment is a medical technique used to estimate how far a child's bones have developed compared to what is expected at their age. This assessment is typically performed using an X-ray of the left hand and wrist, as these regions exhibit predictable and well-documented patterns of bone growth across childhood and adolescence. By examining the shape, size, and level of ossification of individual bones, clinicians can determine the child's "bone age," which may correspond to, exceed, or lag their chronological age (Martin et al., 2011; Halabi et al., 2019; Spampinato et al., 2017).

This process is commonly used in Pediatric endocrinology to estimate skeletal maturity and support the evaluation of growth patterns (Martin et al., 2011). Unlike chronological age, which simply measures the number of years a child has lived, bone age reflects the biological progression of growth, offering a more nuanced view of development (Spampinato et al., 2017). Discrepancies between bone age and chronological age can arise due to a variety of factors, including hormonal imbalances, genetic influences, nutritional status, and certain pathological conditions. By identifying whether skeletal development is advanced, delayed, or on track, bone age assessment helps guide clinical decisions, inform prognosis, and tailor interventions such as growth hormone therapy, management of early or delayed puberty, and monitoring of long-term growth outcomes.

The growing demand for reliable, efficient, and objective methods has made the development of automated bone age assessment increasingly relevant. With rising patient volumes and the expansion of imaging technologies, manual assessments are increasingly difficult to scale in busy clinical environments due to high patient volumes and the need for timely results (Martin et al., 2011; Halabi et al., 2019). Automation promises to address key challenges by reducing variability, increasing speed, and offering standardised results that can be integrated into clinical workflows. This is especially significant in Pediatrics, where accurate and timely diagnosis is crucial to optimise treatment during critical windows of growth and development.

This method is a cornerstone of Pediatric endocrinology and beyond, with applications ranging from the diagnosis of growth and endocrine disorders to the monitoring of therapy, prediction

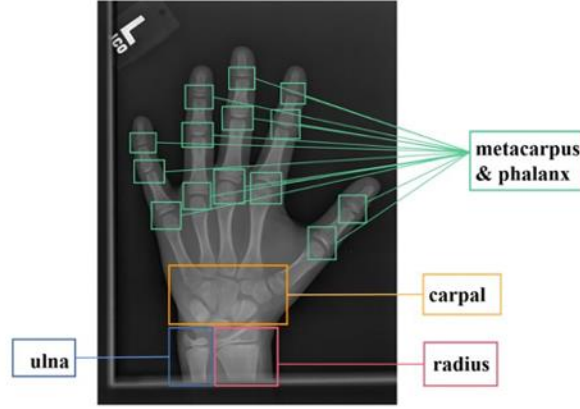
of adult height, and even forensic investigations. While radiographs of the hand and wrist provide a reliable basis for estimating skeletal maturity, traditional manual methods remain limited by population bias, subjectivity, and inefficiency. These limitations provide the foundation for the increasing adoption of computational and automated techniques, including machine learning and deep learning, which aim to deliver objective, scalable, and clinically reliable solutions to a problem that has remained central to Pediatric healthcare for decades.

Doctors usually examine X-rays of the hand and wrist, since these bones follow predictable growth patterns that can be compared against expected milestones. By checking whether a child's bone development is ahead, behind, or in line with their chronological age, clinicians can make better decisions about treatment and predict adult height.

2.2 Traditional Methods for Bone Age Assessment

Historically, bone age estimation has been performed manually using radiographs of the non-dominant hand and wrist. The two most widely used methods are the Greulich and Pyle (G&P) and Tanner-Whitehouse (TW) methods. The G&P method involves comparing a child's radiograph with a series of standard reference images in the Greulich and Pyle atlas, which was derived from a homogeneous North American cohort in the mid-20th century. Although widely used, this method is inherently subjective and limited in its demographic applicability (Bagnall and Davis, 2014).

The G&P method compares the radiographs with the reference atlas until the most similar atlas is selected, and the labelled age of the selected atlas represents the estimated bone age. Compared to the G&P method, the TW method is more complex, it analyzes specific regions of interest (ROIs, which consist of the radius, ulna, carpal, and metacarpus & phalanx bones, as shown in Figure 2.1), and bone age is evaluated individually through a scoring mechanism, rather than based on the entire radiograph. Due to the critical anatomical features in ROIs playing important roles in Bone age assessment, each ROI is assessed by a numerical scoring system, and the final bone age is estimated by averaging all ROI scores (Li et al., 2023).



(Li et al., 2023)

Figure 2.1: ROIs of bone X-ray images for bone age assessment

The TW method, introduced in the 1960s and later refined (TW2, TW3), relies on scoring specific bones based on their developmental stages. It includes the radius, ulna, and small bones (RUS) or carpal bones and uses these scores to calculate a bone age. This approach offers a more quantitative evaluation than G&P but still requires significant expertise and training. Both methods are prone to inter-observer and intra-observer variability making results less reproducible. Moreover, these assessments are time-intensive and not scalable for large clinical loads, prompting the need for automation (Bagnall and Davis, 2014).

In part 1 and part 2 of their seminal papers Martin et al. (2011a, 2011b), comprehensively discussed the clinical applications and limitations of bone age in practice, highlighting the variability in methods and the importance of standardized evaluation protocols. They argued that while traditional methods serve as a foundation, technological augmentation is essential for increased diagnostic reliability, especially with growing digital infrastructure in modern radiology.

2.3 Diverse Datasets used in Bone Age Studies

Several datasets have been employed for bone age estimation across various studies. Thodberg's BoneXpert system was trained using data from the Greulich and Pyle atlas and multiple clinical institutions across Europe and North America (Thodberg et al., 2009). Studies by Caliebe et al. (2013) utilized datasets spanning multiple ethnic groups and clinical scenarios, enabling the identification of demographic differences in bone maturation. These findings underscored the importance of inclusive datasets to mitigate ethnic and gender biases in machine learning systems for bone age assessment.

Caliebe et al. (2013), as cited in, highlighted the discrepancies in bone maturation across populations, suggesting that predictive models must account for such variability. Moreover, Martin et al. (2011, part 1; 2011, part 2) emphasized the impact of endocrinological disorders on BA, requiring datasets that capture pathological as well as normal growth trajectories.

However, one of the key limitations of traditional datasets is their limited accessibility, small size, and lack of standard annotations. Most datasets were not designed for algorithmic learning and lacked uniform image quality, making them suboptimal for training modern machine learning models. Furthermore, these datasets often lacked accompanying metadata such as height, weight, sex, or ethnicity, which could help build multimodal prediction systems.

Other than this, the CQJTJ dataset is an additional dataset collected by the radiology department of Chongqing Jintongjia Children's Hospital. It consists of 3,551 left-hand X-ray images and corresponding clinical reports. Among these images, some are collected from the same subjects at different periods, 1,502 images are obtained from male subjects and 1,949 images are obtained from female subjects, with an age range of 13–218 months, mainly for children aged 4–16 years. This dataset is not publicly available, it is collected by the radiology department of Chongqing Jintongjia Children's Hospital and used exclusively for internal research purposes in the relevant study (Li et al., 2023).

2.5 Deep Learning approach

Deep learning, particularly Convolutional Neural Networks (CNNs), has transformed the field of medical image analysis, and bone age assessment is no exception. Unlike traditional methods that depend on handcrafted features, CNNs can learn hierarchical features directly from raw radiographs. This means that rather than a human expert defining which aspects of the bone to measure, the model progressively extracts edges, shapes, and higher-level skeletal patterns across its layers, making it particularly suitable for detecting the subtle variations in ossification and growth plate development that are crucial in bone age estimation.

2.5.1 Convolutional Neural Network

A CNN is composed of several layers, each performing a specific role in transforming the input image into a meaningful representation. The early layers typically consist of convolutional filters that slide across the image, detecting low-level features such as edges and curves. As depth increases, the network learns more abstract features, such as bone contours or joint structures. Pooling layers are used to reduce spatial dimensions, making the network more

efficient while preserving the most important information. Deeper layers capture increasingly complex patterns until a fully connected layer maps these features to a final output.

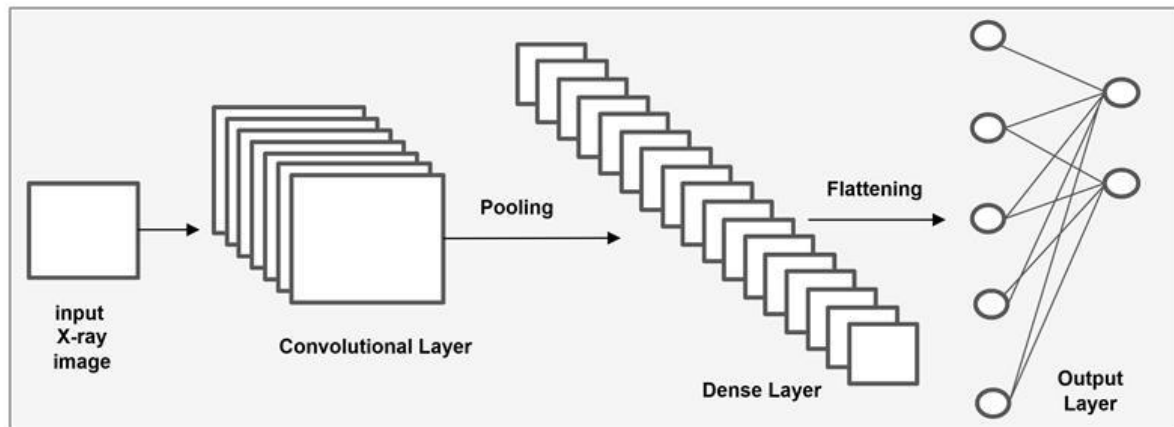


Figure 2.2: CNN Architecture

Convolutional layer

The convolutional layer is responsible for extracting features from the input images. It applies a set of learnable filters, also called kernels, which are small matrices commonly 2×2 , 3×3 , or 5×5 in size. These kernels slide across the image and perform a dot product between their weights and the corresponding patch of the image. The result is a set of feature maps, which highlight important patterns such as edges, textures, or shapes. For example, if 12 filters are used in a convolutional layer, the output will be a volume of size $32 \times 32 \times 12$, where each of the 12 channels corresponds to the features detected by one filter. This is a basic example with a 2×2 kernel:

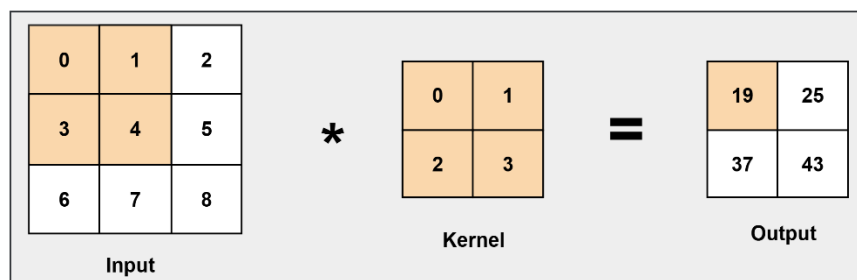


Figure 2.3: Convolution Operation

Pooling layer

A pooling layer is periodically added to reduce the spatial size of the feature maps. Its main purpose is to make computation faster, reduce memory usage, and help prevent overfitting by simplifying the learned representations. The two most common types are max pooling and

average pooling. For example, applying max pooling with a 2×2 filter and a stride of 2 would shrink the feature map, so if the input volume was $32 \times 32 \times 12$, the output would become $16 \times 16 \times 12$. Example of max pooling.

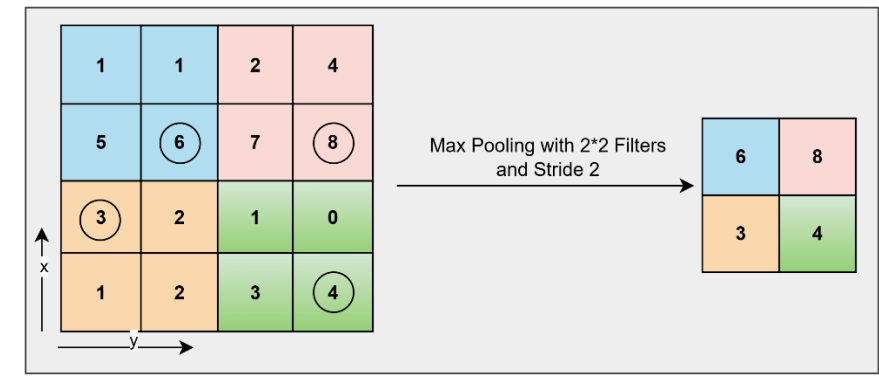


Figure 2.4: Max Pooling

Fully connected layer

After successive rounds of convolution and pooling, the resulting feature maps are flattened and passed into a fully connected layer (FC layer). Unlike convolutional layers, which operate locally, the FC layer integrates information globally, combining all the learned features into a single predictive output. For bone age assessment, this output is typically a continuous value representing the estimated skeletal age in months. In this way, the FC layer serves as the decision-making stage of the CNN, translating extracted patterns into clinically interpretable predictions.

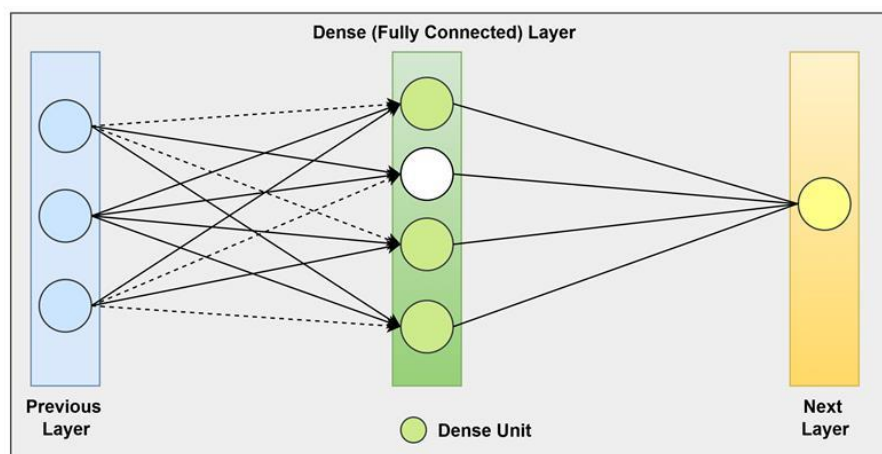


Figure 2.5: Fully Connected Layer

2.5.2 Activation function

Activation functions introduce non-linearity into CNNs, enabling them to model complex relationships. The Rectified Linear Unit (ReLU), defined as $f(x) = \max(0, x)$, is widely used due to its simplicity and ability to mitigate the vanishing gradient problem during backpropagation. Other functions, such as sigmoid and tanh, were common in early neural networks but are less effective for deep architectures due to saturation issues. In medical imaging, these activation functions ensure that CNNs can capture the non-linear developmental differences present in children of varying ages and genders.

2.5.3 Loss Function

The mean absolute error (MAE) loss is employed for regression tasks, measuring the average absolute difference between predicted and actual bone ages. This loss penalizes errors linearly, ensuring consistent treatment of deviations regardless of their magnitude, which promotes robust model performance. Alternative losses, such as mean squared error (MSE) or Huber loss, are also considered, with MAE chosen for its simplicity and suitability for the continuous nature of bone age predictions. This section analyzes the mathematical properties and training implications of these loss functions.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Taken together, the CNN architecture provides a powerful framework for automated bone age prediction. Its ability to automatically learn relevant features from radiographs, coupled with optimisation techniques, makes it a good alternative to manual methods and early feature engineered systems. By combining convolutional layers for feature extraction, pooling for efficiency, fully connected layers for decision making, and carefully chosen activation and loss functions, CNNs deliver accurate, scalable, and clinically interpretable predictions that align with the goals of modern Pediatric endocrinology.

2.6 Advantages of Deep Learning-Based Bone Age Assessment

The application of deep learning to bone age prediction brings several key advantages like the models provides consistent outputs, minimizing the subjective variability seen in human assessments. This consistency is particularly valuable in high-throughput clinical environments or in regions lacking experienced radiologists.

Once deployed, these models can generate predictions in seconds, significantly reducing the time burden on clinicians and improving throughput, especially in hospitals dealing with large Pediatric populations. Models trained on well-arranged datasets can achieve performance metrics on par with or exceeding that of human experts, especially when ensemble models or advanced architectures are used. Additionally, these models can not only predict bone age but also flag anomalies, highlight discrepancies, or recommend further investigations, thereby augmenting clinical decision-making. Martin et al. (2022) and Halabi et al. (2019)

Bone age assessment remains a foundation of the management of Pediatric growth disorders, with traditional methods such as G&P and TW having served clinicians for decades. However, these manual approaches are increasingly limited by their subjective nature, inter-observer variability, and inefficiency. Recent advances in machine learning, especially deep learning, have provided robust solutions to these limitations.

Research led by Martin et al. (2011, 2022), Halabi et al. (2019), and others has shown that CNNs can outperform traditional models and match clinical experts in bone age prediction accuracy. Integrating these models into user-friendly platforms enables clinicians to access accurate, real-time, and scalable assessments without compromising diagnostic integrity. By combining clinical relevance, computational strength, and practical application, these technologies pave the way for a more efficient, equitable, and data-driven approach to Pediatric care.

Chapter 3: Design

The Design of this project refers to the structured plan and workflow implemented to develop an automated system for predicting skeletal age from pediatric hand X-ray images. The system employs a deep learning approach to automatically extract relevant features from X-ray images and predict bone age as a continuous variable. This section outlines the overall workflow of the proposed framework, describing how raw images and metadata are processed, analyzed, and passed through the CNN model, followed by postprocessing steps to generate clinically meaningful predictions.

3.1 System Architecture

The system architecture of this framework is designed to provide an end-to-end, real-time solution suitable for clinical use. The pipeline begins with pediatric hand X-ray images as input, which are either uploaded through a graphical user interface (GUI) or retrieved from a local database. These raw images first undergo a preprocessing stage. This ensures that skeletal structures are standardized for input into the deep learning model.

Following preprocessing, the images are passed through a convolutional neural network (CNN), which serves as the core predictive engine. The CNN automatically extracts hierarchical features, ranging from low-level edges and textures to high-level anatomical patterns, and outputs an estimated skeletal age in months. The model is trained to minimize prediction error, producing reliable and fine-grained age assessments that are clinically relevant (Wu, 2017).

Finally, the predicted bone age is presented through a user-friendly GUI, which allows clinicians to visualize the X-ray alongside the predicted age and optionally save or export the results. The architecture is optimized for real-time performance, ensuring minimal latency between image input and prediction output. By integrating preprocessing, CNN-based prediction, and an interactive interface, the system provides a seamless, accurate, and efficient tool for supporting pediatric growth assessment in clinical settings.

3.2 Input / Data Loading

The pipeline begins with the input stage, where pediatric hand X-ray images are obtained from the RSNA Bone Age 2017 dataset, which serves as the primary source for model development (Halabi et al., 2019). This dataset consists of 14,236 X-ray images divided into 12,611 training

images, 1,425 validation images, and 200 test images. Each image is stored in PNG format and varies in resolution. Alongside the images, metadata is provided for each sample, including the patient's skeletal age in months and gender. Gender information is particularly important, as skeletal growth differs between males and females, and the inclusion of this metadata allows the model to account for such variations. The availability of a large number of labeled images makes this dataset highly suitable for training deep learning models, enabling them to learn complex patterns in bone structure and development.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to gain an in-depth understanding of the dataset before model training. This step involves examining the distribution of bone ages in both the training and validation sets, including calculating statistical measures such as mean, median, and skewness to identify potential imbalances or biases. Skewness analysis helps to understand the shape and symmetry of data distribution, as it indicates whether values are evenly spread around the mean or lean toward one side. A left-skewed (negative skew) distribution means most data values are concentrated on the higher end, with a longer tail extending toward smaller values, while a right-skewed (positive skew) distribution has most values concentrated on the lower end, with a longer tail extending toward larger values. Gender distribution is also analyzed to ensure adequate representation of both male and female subjects, reducing the risk of gender-related bias in predictions. In addition to statistical analysis, sample X-ray images are visually inspected to assess image quality, orientation, and resolution variations, providing a qualitative understanding of the data. Insights gained from EDA help guide preprocessing steps and inform model design decisions, ensuring that the dataset is clean, representative, and suitable for training a reliable predictive model.

3.4 Preprocessing Techniques

Preprocessing in machine learning and deep learning refers to the set of techniques applied to raw data to prepare it for modeling, ensuring that it is clean, consistent, and in a format suitable for training algorithms. This step is essential because raw data often contains noise, missing values, irrelevant features, or variations in scale and format that can negatively affect model performance. In image-based tasks, preprocessing may include resizing, normalization, filtering, or contrast enhancement to standardize inputs and highlight relevant features. In tabular or text data, it may involve handling missing values, encoding categorical variables, or scaling numerical features. Proper preprocessing improves model convergence, reduces

training time, prevents overfitting, and allows the algorithm to focus on learning meaningful patterns rather than being distracted by inconsistencies or irrelevant variations in the data.

3.4.1 Adaptive CLAHE (Contrast Limited Adaptive Histogram Equalization)

CLAHE is a technique that enhances the local contrast of an image by dividing the image into smaller parts and adjust the contrast in each part. The clip limit prevents over-amplification of noise. It is widely used in medical imaging to highlight fine structural details without introducing artifacts.

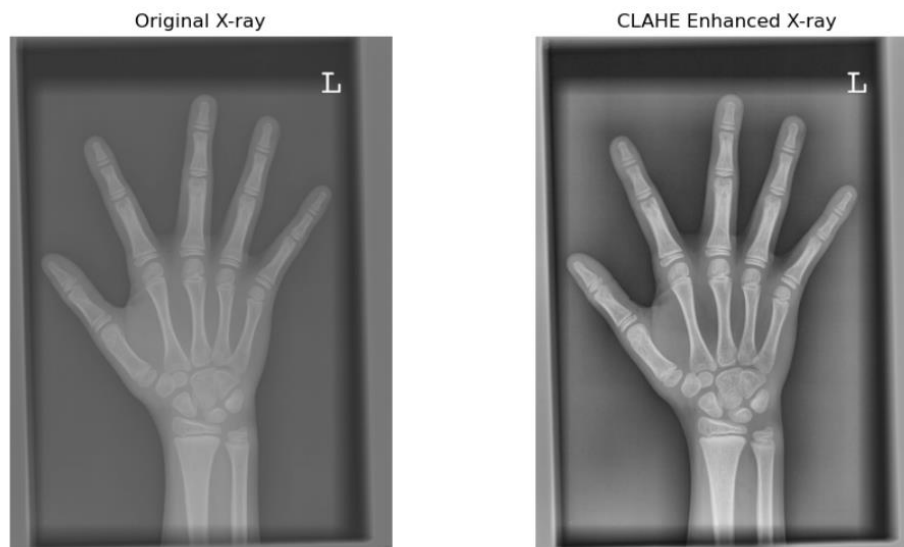


Figure 3.1: CLAHE

3.4.2 Edge-Preserving Bilateral Filtering

Bilateral filtering smooths an image to reduce noise while preserving sharp edges. It combines both spatial proximity and pixel intensity similarity to selectively blur regions, maintaining important structural boundaries, such as bones in X-ray images.

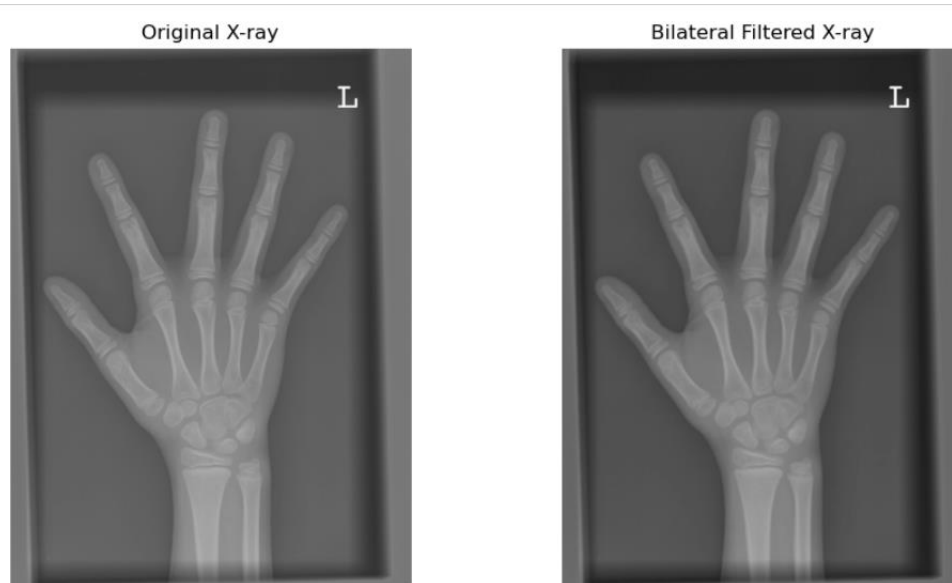


Figure 3.2: Bilateral Filtering

3.4.3 Histogram Equalization

Histogram equalization adjusts the image intensity distribution to fully utilize the available pixel range. This improves contrast uniformly across the image and corrects for differences in exposure or lighting conditions, making features more distinguishable.

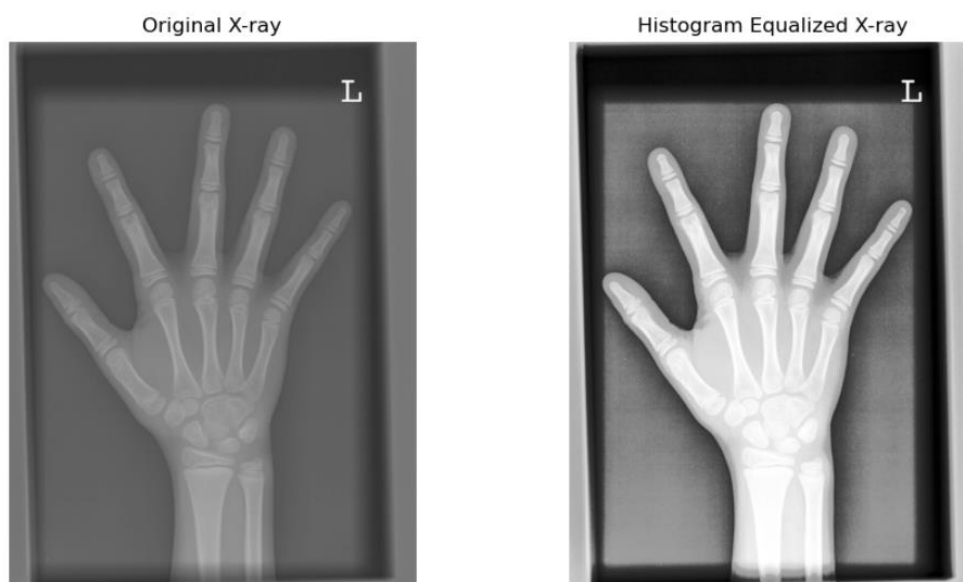


Figure 3.3: Histogram Equalization

3.4.4 Resizing

Resizing adjusts all images to a consistent dimension using interpolation techniques. This ensures that every image has the same input size for neural networks, which is critical for batch processing and consistent feature extraction.

3.4.5 Normalization

Normalization scales pixel intensity values to a standard range (e.g., [0,1] or [0,255]) and may standardize based on dataset-specific mean and standard deviation. This reduces numerical variation between images, improves network convergence, and stabilizes training.

3.5 Data Augmentation

Data augmentation is a process used in machine learning and deep learning to artificially increase the size and diversity of a dataset by applying a set of transformations to the original data. These techniques help improve model generalization, reduce overfitting, and make the model robust to variations in real-world data (Ornek and Ceylan, 2019). In this study, augmentation was applied on the training set to introduce controlled variability while preserving the biological correctness of hand anatomy. Only the techniques that proved effective in this project were used.

3.5.1 Horizontal Flipping

Creating a mirror image of the original X-ray along the vertical axis. This increases dataset diversity and helps the model generalize to left–right orientation variations.

3.5.2 Rotation Range

Rotating the image by a small random angle within a specified range ($\pm 10^\circ$). This simulates differences in imaging angles and improves the model’s tolerance to orientation variability.

3.5.3 Zoom and Cropping

Randomly cropping a portion of the image and resizing it to the desired input size. This acts as a zoom augmentation, enabling the model to learn scale-invariant features and improving robustness to changes in imaging distance.

3.5.4 Rescaling / Normalization

Scaling pixel values to a standard range (e.g., $[0,1]$) and adjusting them using dataset-specific mean and standard deviation values. This standardizes intensity distributions, stabilizes training, and improves model convergence.

3.6 Additional Augmentation Techniques

Other augmentation strategies could be explored to further enhance dataset diversity, though they were not applied in this study:

Brightness Adjustment: Randomly modifying brightness could simulate variations in X-ray exposure and improve generalization.

Contrast Adjustment: Adjusting contrast may enhance the visibility of skeletal structures in under- or overexposed images.

Elastic Transformations: Minor elastic deformations could help the network learn invariant skeletal features, provided anatomical correctness is preserved.

These optional augmentations were not included in the current study, as the selected techniques flipping, rotation, zoom/cropping, and normalization were sufficient for achieving robust model performance.

3.7 Deep Learning Models

In this study, four different CNN architectures ResNet, DenseNet, MobileNet, and AlexNet were implemented and compared to evaluate their effectiveness in pediatric bone age prediction. The goal was to identify which model delivers the best balance of accuracy, efficiency, and generalization for Xray image analysis (Nguyen et al., 2023).

3.7.1 ResNet (Residual Network)

ResNet is a deep convolutional neural network architecture that was introduced to overcome the limitations of training very deep networks, particularly the vanishing gradient problem. In normal deep networks, as layers increase, the training signal (gradients) becomes very small when going backwards, making it hard for earlier layers to learn. ResNet solves this issue using skip connections, which allow the network to learn identity mappings. These connections jump one or more layers by adding the input of a layer directly to its output, enabling gradients to flow more easily during training. This design allows ResNet to train hundreds of layers without

degradation in performance, making it highly suitable for extracting complex hierarchical features from medical images (He et al., 2016).

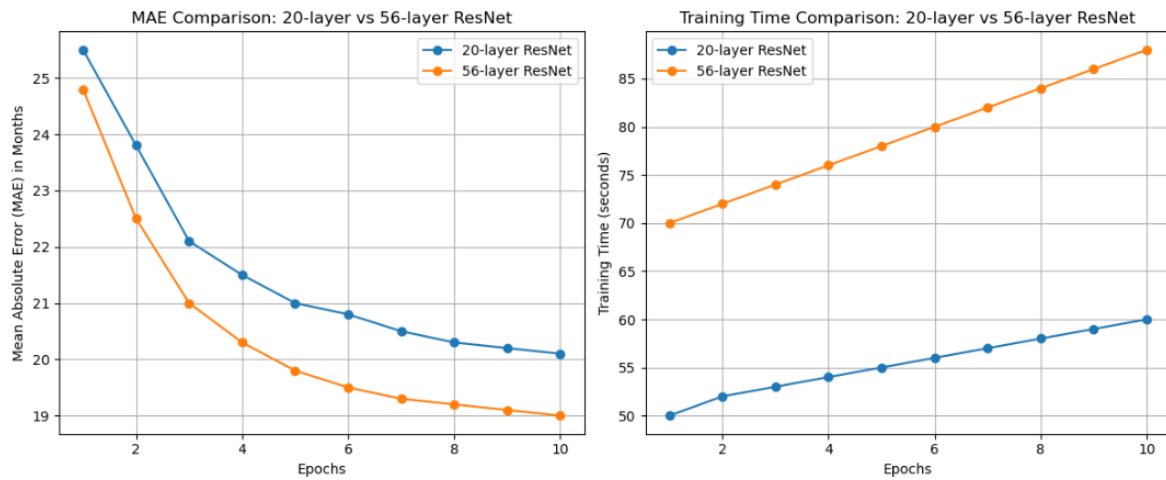


Figure 3.4: 20 vs 50 Layer ResNet

The typical architecture of ResNet includes the following components: Input Layer, Initial Convolutional Layer, Max Pooling Layer, Residual layer and Global Average Pooling (GAP)

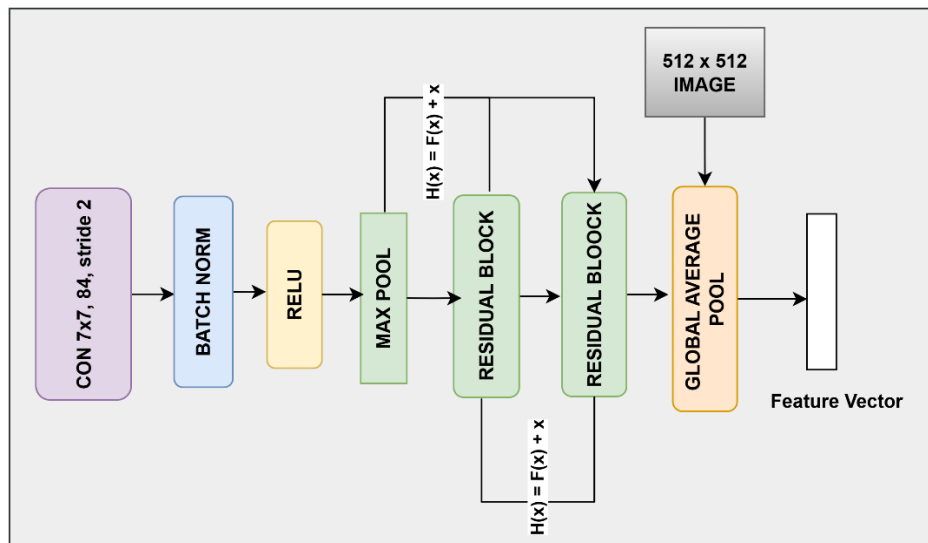


Figure 3.5: ResNet Architecture

ResNet is used in this project because it can learn very detailed patterns in hand X-rays that are important for predicting bone age. Its skip connections make it easier to train very deep networks without the problem of vanishing gradients, so the model learns more effectively. ResNet can pick up both small details like edges and textures, as well as bigger patterns in bone structure, which is important since even tiny changes in bone growth matter for age prediction.

Another advantage is that ResNet is flexible — it can be easily adjusted from classification tasks to regression, which makes it a good fit for predicting skeletal age as a continuous value.

3.7.2 DenseNet (Densely Connected Convolutional Network)

DenseNet is a deep learning architecture that improves feature propagation and reuse by introducing dense connections between layers. Unlike traditional CNNs where each layer passes information only to the next, DenseNet connects each layer to every subsequent layer in a feed-forward fashion. This means the feature maps learned by earlier layers are directly available to later layers, which encourages feature reuse and reduces redundancy. As a result, DenseNet can achieve better accuracy with fewer parameters compared to other deep architectures. Given an L layer network, a traditional convolutional network would consist of L connections. Now with the DenseNet, for each layer, exists direct connections connecting to all subsequent layers, therefore resulting in $L(L+1)/2$ connections (Huang et al., 2017)

The typical architecture of DenseNet includes: Dense Blocks, Bottleneck Layers and Transition Layers.

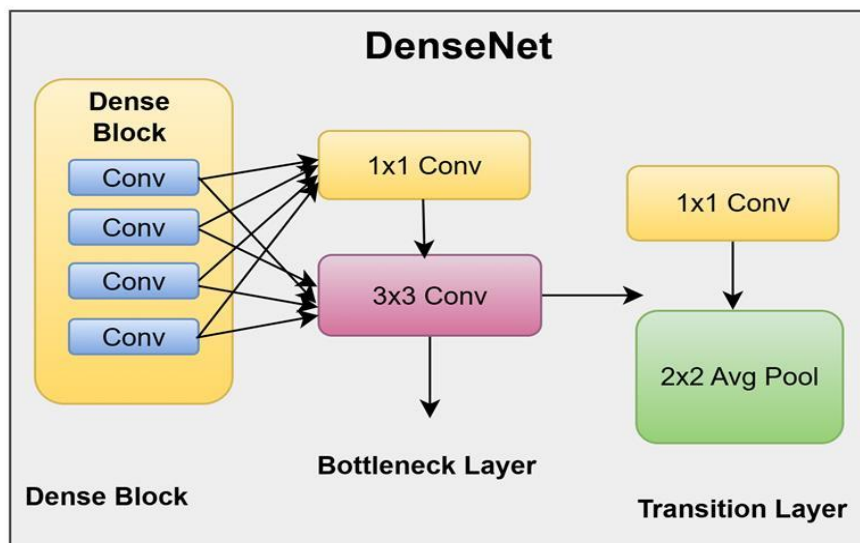


Figure 3.6: DenseNet Architecture

DenseNet comes in several variants, distinguished primarily by their depth and number of layers (Huang *et al.*, 2017).

Model	top-1	top-5
DenseNet-121 (k=32)	25.02 (23.61)	7.71 (6.66)
DenseNet-169 (k=32)	23.80 (22.08)	6.85 (5.92)

DenseNet-201 (k=32)	22.58 (21.46)	6.34 (5.54)
DenseNet-161 (k=48)	22.33 (20.85)	6.15 (5.30)

Table 3.1: DenseNet Variants

DenseNet architecture is particularly well-suited for this project because its dense connections encourage feature reuse, making the model more parameter-efficient and less prone to overfitting. These connections also ensure strong gradient flow during backpropagation, which makes training easier and more stable. This design is especially beneficial for medical imaging tasks, where subtle skeletal features are critical, as it allows earlier low-level features such as edges and textures to remain directly accessible to deeper layers, enhancing the model’s ability to capture fine-grained anatomical details.

3.7.3 MobileNet

MobileNet is a family of lightweight deep convolutional neural network (CNN) architectures designed specifically for efficient deployment on mobile and embedded devices with limited computational resources, such as smartphones, IoT devices, and edge hardware. Introduced by Google in 2017 (Howard et al., 2017), MobileNet architectures prioritize low latency, reduced memory footprint, and energy efficiency while maintaining competitive accuracy for tasks like image classification, object detection, and segmentation. This is achieved with depthwise separable convolutions, a key innovation that significantly reduces the number of parameters and computational operations compared to traditional CNNs like ResNet or VGG. For this project, MobileNetV2 was chosen for its lightweight design, allowing fast bone age predictions in clinics, and it includes gender to help achieve an MAE of about 21 months.

The MobileV2 Architecture: Depthwise Separable Convolutions, Depthwise Convolution, Pointwise Convolution, Inverted Residuals, Linear Bottlenecks, Width and Resolution Multipliers:

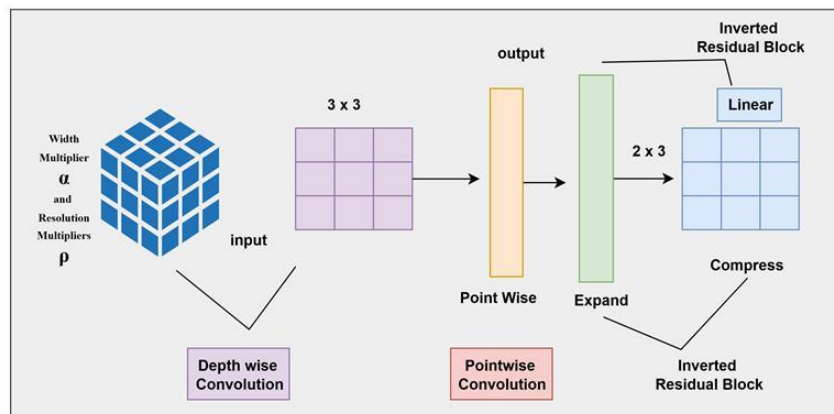


Figure 3.7: MobileNet Architecture

MobileNet V2 is a powerful and efficient neural network architecture designed for mobile and embedded applications. Whether for image classification, object detection, or other vision-based tasks, MobileNet V2 provides a robust solution for deploying AI on resource-constrained devices.

3.7.4 AlexNet

AlexNet is one of the earliest and most influential deep CNN architectures, which demonstrated the power of deep learning in computer vision by winning the ImageNet challenge in 2012. It consists of multiple convolutional layers for feature extraction followed by fully connected layers for classification or regression. AlexNet introduced innovations like ReLU activation for faster training, dropout for regularization, and data augmentation to reduce overfitting. While relatively shallow compared to modern networks, AlexNet serves as a strong baseline model for medical image analysis (Alom et al., 2018)

The typical structure of AlexNet includes: Convolutional Layers, Pooling Layers, ReLU Activation, Dropout Layers and Fully Connected Layers.

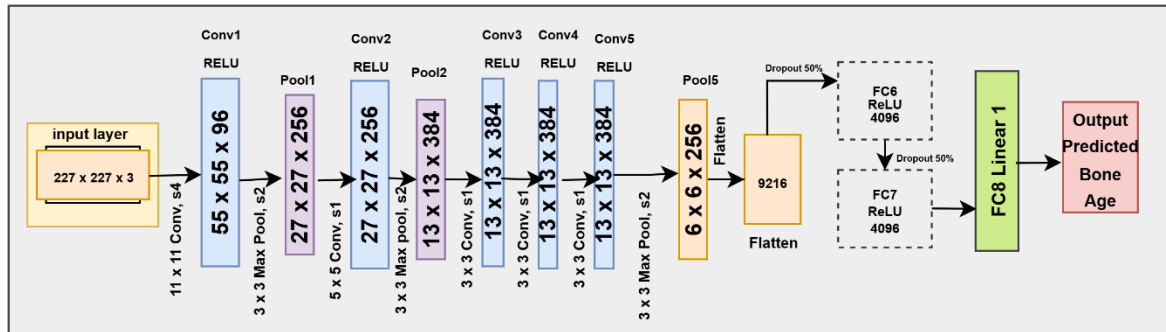


Figure 3.8: AlexNet Architecture

AlexNet serves as a baseline model to compare against deeper and more advanced architectures. Model Demonstrates the importance of regularization (dropout) and non-linear activations (ReLU), which remain essential in modern architectures.

Although not as deep or efficient as ResNet or MobileNet, AlexNet provides valuable insights into how simpler architectures perform on medical imaging tasks.

3.8 GUI Design

Streamlit was chosen for building the graphical user interface due to its low-code framework, which enables rapid development of interactive applications. Its simplicity minimizes the need for complex backend setup and allows greater focus on the model development process. The platform's built-in interactive widgets make it possible for users to upload medical images, provide metadata, and view results in an accessible environment. The interface was designed with usability in mind, incorporating features such as visualization of original and preprocessed images, display of predicted bone age, and options to generate downloadable PDF reports for documentation. These design choices aimed to create a system that is intuitive for non-technical users and adaptable for research and potential clinical exploration. Fig 3.9 shows streamlit architecture for this project.

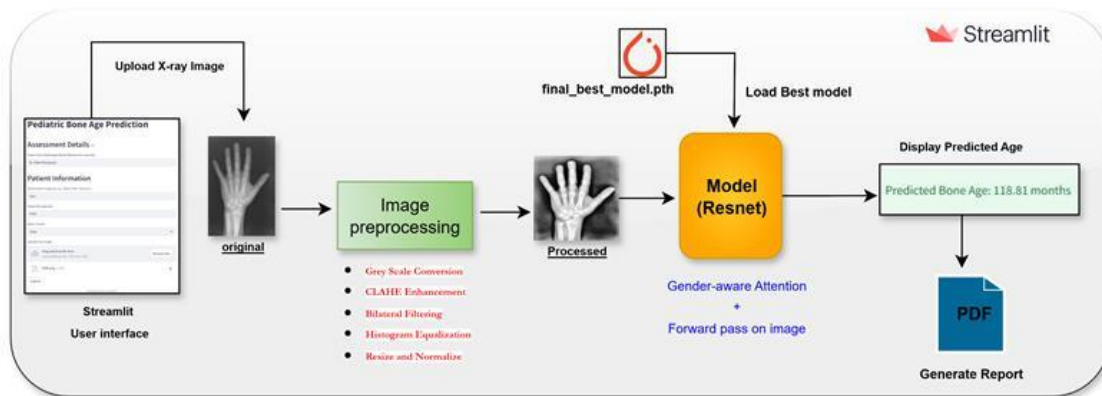


Figure 3.9: GUI Architecture

3.9 Computational Requirements

Kaggle was used as the primary computational platform, offering an accessible balance of GPU resources, storage, and ease of use for large-scale model training. With free access to NVIDIA T4/P100 GPUs for up to 30 hours per week, it supported efficient experimentation without the need for dedicated hardware. The platform also provided 100GB of persistent storage, which was useful for maintaining preprocessed datasets, intermediate results, and trained models. Its dataset and notebook interface further simplified file management and remote access.

For local development and testing, the setup required hardware with at least 16GB RAM, an NVIDIA GPU compatible with CUDA 11.8, and a minimum of 100GB storage. The software

environment was based on Python (3.8+), with PyTorch for model development, torchvision and OpenCV for preprocessing, Streamlit for deployment, and ReportLab for report generation. Dependencies were tracked through a requirements file to maintain consistency across platforms. Visual Studio Code was selected as the primary development environment for its lightweight design, integrated debugging, and strong Python support.

To provide a clear overview of the end-to-end pipeline, a flowchart is presented. This visual representation illustrates the sequential steps, from image input to GUI output, helping to understand the system's workflow and data flow. The diagram serves as a useful reference for both technical and clinical audiences.

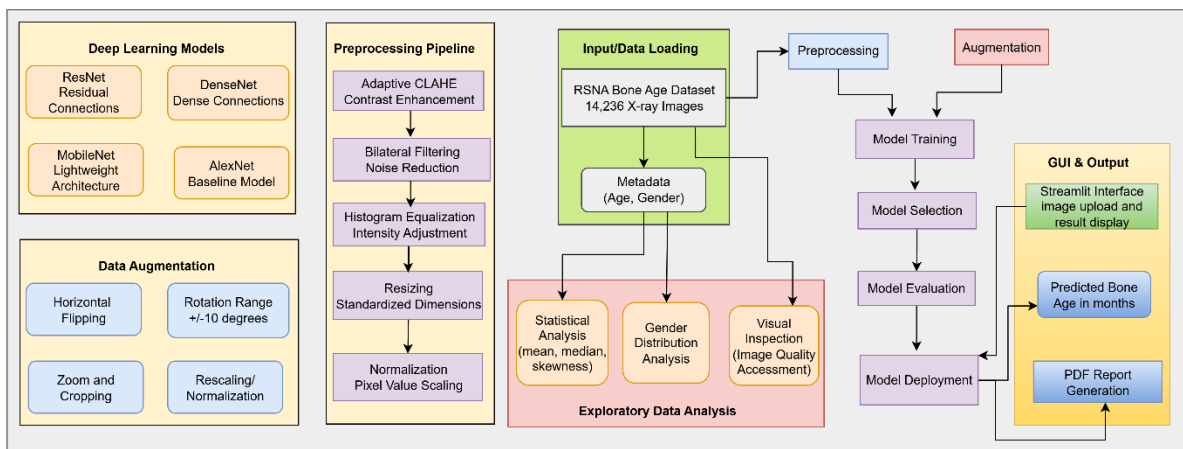


Figure 3.10: System Design

3.10 Output

The final output of the system is the estimated skeletal age in months for each input X-ray image. These predictions can be used by clinicians to assess pediatric growth patterns, detect developmental abnormalities, and guide treatment decisions. By providing an automated and reliable estimation of bone age, the system reduces the time and effort required for manual assessment, while maintaining high accuracy and consistency across large datasets. The combination of preprocessing, deep learning-based prediction, and optional postprocessing ensures that the pipeline produces robust and clinically valuable results.

Chapter 4: Implementation

This section outlines the development and integration of an end-to-end pipeline for automated Pediatric bone age assessment using the RSNA dataset. The aim of the implementation is to build a reproducible and modular workflow that systematically processes raw X-ray images, applies preprocessing and enhancement techniques, trains deep learning models, and delivers predictions through a user-oriented interface. The system follows a structured sequence of stages: data loading and cleaning, exploratory data analysis (EDA), image preprocessing, , and final output generation. Each stage was designed to ensure consistency of input data, robustness of model training, and clarity of predictions, supporting both technical performance and clinical relevance.

To enable this pipeline, a range of libraries was imported to support the different stages of implementation. Pandas, NumPy, and PIL were employed for data loading and exploration. OpenCV was applied to conduct preprocessing of the image dataset, while Torch facilitated data transformation, model construction, and training. Seaborn and Matplotlib were utilised for data visualisation and evaluation of results. Additionally, Pickle was used to save the best-performing model for future use. The inclusion of these libraries ensured the availability of essential functionalities required for data preparation, model development, and evaluation.

4.1 Loading and Cleaning Dataset

The implementation process began with dataset preparation, using the RSNA Pediatric Bone Age dataset as the foundation for analysis (Halabi et al., 2019). It comprises annotated radiographs organized across three main folders: a training set containing 12,611 images, and two validation folders, val1 contains 800 images and val2 has 625 images. Each image is associated with metadata, including an image ID, bone age in months, and a gender label. These annotations were provided in CSV files corresponding to the respective datasets.

As part of initial cleaning, a new column “path” was created to store the full file paths corresponding to each X-ray, enabling straightforward image loading during model training. Column names were standardized across datasets: in the training set, male was renamed to Gender and its Boolean values were converted to categorical labels (Male and Female), while in the validation set, Image ID and Bone Age in months were similarly renamed to match the training set. The columns in both datasets were then reordered to a uniform structure id, boneage, Gender, and path.

4.2 EDA - Exploratory Data Analysis

The exploratory data analysis phase provided an initial understanding of dataset composition. The process began with checking the overall dataset shape and identifying whether any missing values were present in the annotation files. Dataset statistics were then summarised, and distributions were examined to gain an overview of the data composition. The summary of the dataset shows a mean bone age of approximately 127.3 months with a standard deviation of 41.2, reflecting a broad variability in age distribution. The median bone age is 132 months, while the interquartile range spans from 96 to 156 months, covering the central 50% of observations within the overall range of 1 to 228 months.

Further investigated bone age distribution and gender distribution providing insight into sample representation. The distribution of bone ages and gender composition are presented in Figures X and Y, respectively.

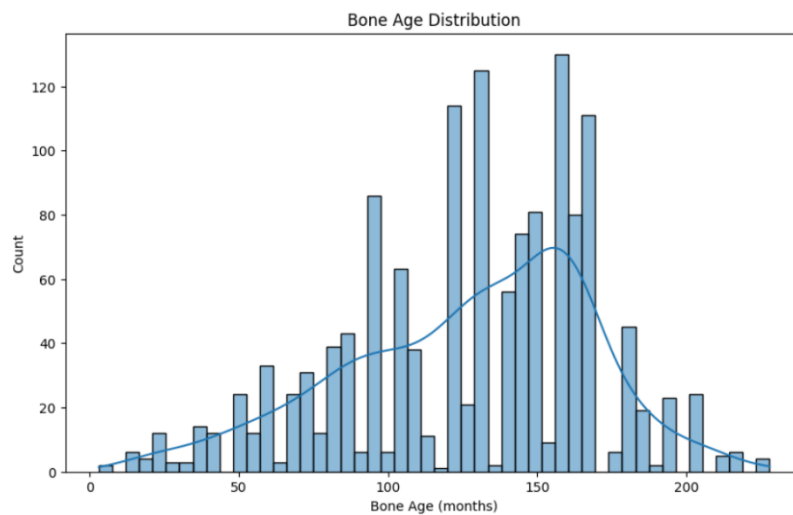


Figure 4.1: Distribution of bone ages across the dataset (in months).

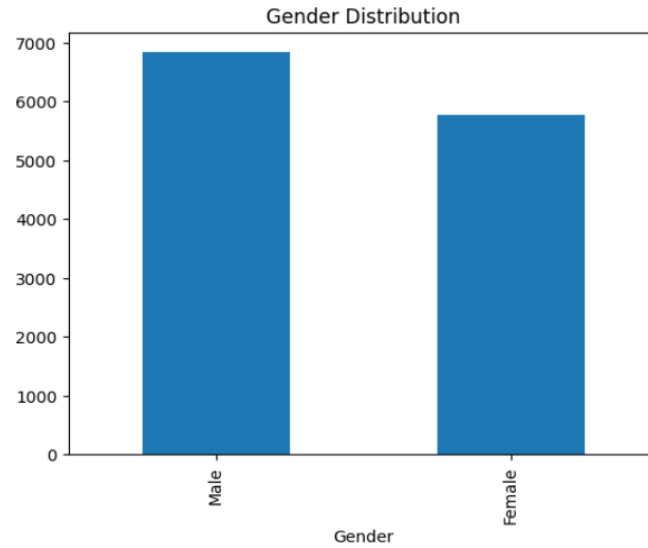


Figure 4.2: Gender distribution of the dataset (male vs. female).

The dataset consists of 54.2% male and 45.8% female subjects, highlighting a predominance of male samples.

4.2.1 Skewness

Skewness within the dataset was further analyzed to assess demographic balance, particularly the male-to-female ratio across both training and validation sets. Both datasets exhibited a slight left skew (-0.47), indicating that most bone age values are concentrated toward higher ages, with a longer tail extending toward lower ages. The results provided clarity on how the dataset was structured in terms of demographic balance.

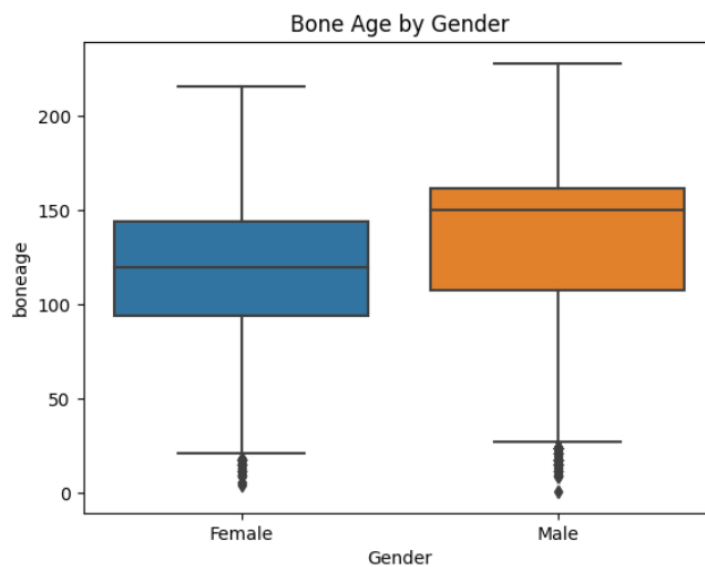


Figure 4.3: Bone age distribution by gender

These steps formed the initial phase of data exploration, followed by additional analyses to further understand dataset characteristics.

4.2.2 Statistical Analysis: T-Test

The analysis also extended to examining statistical differences across groups. A boxplot and t-test were applied to investigate bone age variation by gender. The results indicated that males had an average bone age of 135.3 months, while females had an average of 117.9 months, reflecting a mean difference of approximately 17 months. The t-test confirmed this difference as statistically significant ($t = 24.2$, $p < 0.001$). This finding highlighted the role of gender in bone age distribution, which needed to be considered in subsequent modelling stages.

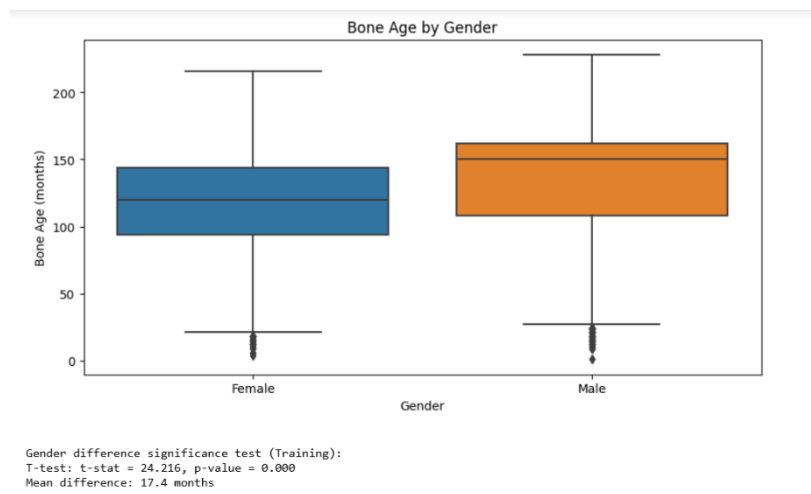


Figure 4.4: Statistical Analysis: T-Test

4.2.3 Bins(age wise)

To improve interpretability and assess dataset balance across developmental stages, bone age values were grouped into discrete bins representing clinically meaningful growth phases (0–24, 25–48, 49–72, 73–96, 97–120, 121–144, 145–168, 169–192, and 193–228 months). This binning process provided a clearer view of how samples were distributed across developmental stages, rather than relying on raw values alone. By applying this procedure to both training and validation sets, consistency and comparability across datasets were maintained.

4.2.4 Duplicate Check

As part of validation, the dataset was checked for duplicate rows and repeated image IDs within the training and validation sets. This ensured the dataset was clean and free from redundancy. The absence of duplicate entries confirmed data integrity and prevented potential biases or errors during training.

4.2.5 Image Exploration

To further understand the dataset, visual inspection of sample images from the training and validation are performed. This process confirmed that the annotation paths correctly matched the corresponding image IDs and provided an opportunity to evaluate the quality of the X-ray images. Reviewing these samples also guided the choice of preprocessing steps required to improve image clarity and ensure suitability for model training. The Fig. Z shows the radiographs from training and validation sets, after the path has been concatenated.

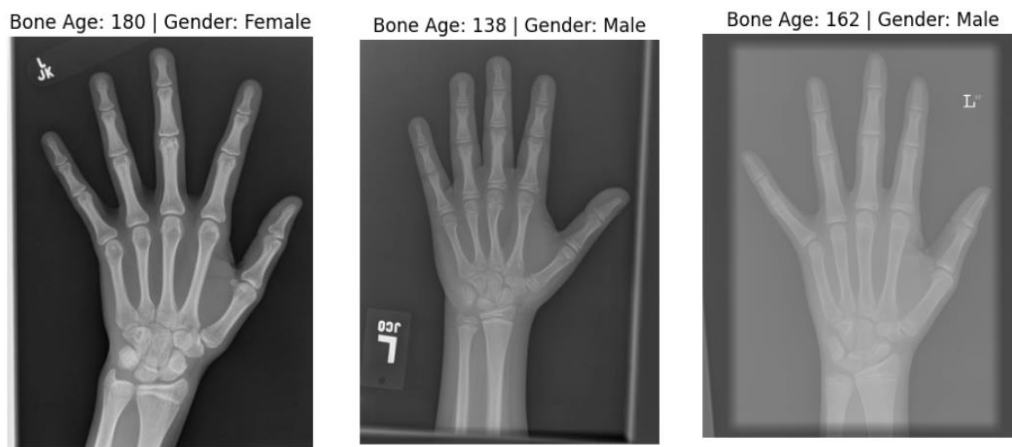


Figure 4.5: X-ray images in the dataset

4.2.6 Height and Width of Images

The dimensions of the X-ray images were analysed to assess variability in height and width. Checking image dimensions ensures that all X-ray images have consistent sizes for model training, helps preserve bone structures during resizing, prevents memory or training issues, and identifies any corrupted or unusual images in the dataset. The fig Z shows height and width of images available.

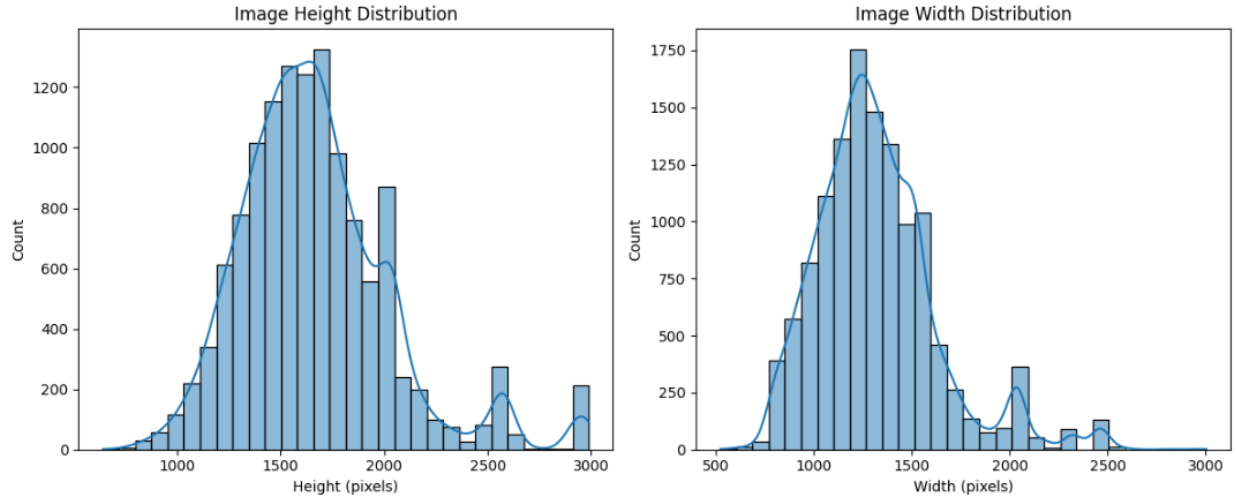


Figure 4.6: Image Dimensions

The output indicates variability and highlights the need for standardization to ensure consistent input dimensions for model training.

4.3 Preprocessing

The preprocessing stage was applied to standardize and enhance the quality of X-ray images before model training. Each image in the dataset was processed using a dedicated function and saved in a separate folder labelled enhance to distinguish it from the raw data. The preprocessing pipeline consisted of several steps designed for clinical bone X-ray images.

Adaptive CLAHE (Contrast Limited Adaptive Histogram Equalization) was applied to improve local contrast across the images. A bilateral filter was used to reduce noise while preserving edge structures, and histogram equalization was applied to normalize intensity distributions between images. Images were resized using a method that maintained the original aspect ratio to prevent distortion of structural features, and intensity normalization was performed to scale pixel values uniformly. Validation and test sets were filtered according to available image IDs to ensure consistency and prevent errors from missing or mismatched files.

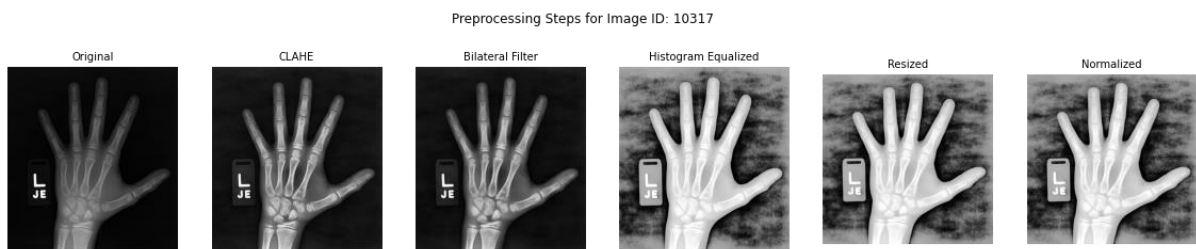


Figure 4.7: Preprocessing steps

Attention was also given to dataset integrity by filtering validation and test sets according to available image IDs, which maintained consistency and prevented issues due to missing or mismatched files. Summary statistics were generated to verify that all images were successfully processed, with an increase in mean intensity values across the enhanced dataset.

These preprocessing steps addressed variability commonly present in clinical imaging, such as differences in illumination, resolution, and noise. Standardization and enhancement ensured uniform input data for subsequent model training and analysis. (Ornek and Ceylan, 2019).

4.3.1 Data Creation and Normalization

In this step, the dataset was prepared to ensure stable and reliable model training. Bone age values were normalized using the mean and standard deviation from the training data, placing all ages on a common scale and facilitating smoother convergence during learning. A custom dataset class was developed to manage image and label loading, while also checking for missing files to avoid downstream errors. To maintain consistency across images, brightness and contrast levels were calculated (mean ≈ 0.51 , standard deviation ≈ 0.28) and applied for normalization. For the training set, basic augmentations such as small rotations, flips, and zooms were introduced to improve generalization and reduce overfitting, whereas the validation and test sets were only resized and normalized to ensure unbiased evaluation. Efficient data loaders were then implemented to batch, shuffle, and feed images into the model, optimizing training speed and stability.

4.3.2 Preprocessing Output

Prior to model training, original X-ray images were visually compared with their enhanced versions to assess the impact of preprocessing. Selected samples were displayed side by side, with raw scans alongside the enhanced images, annotated with bone age and gender information. The enhanced images demonstrated improved contrast, clearer bone structures, and reduced noise, while preserving essential diagnostic details. This comparison illustrated that the preprocessing pipeline produced standardized and high-quality inputs suitable for deep learning models.

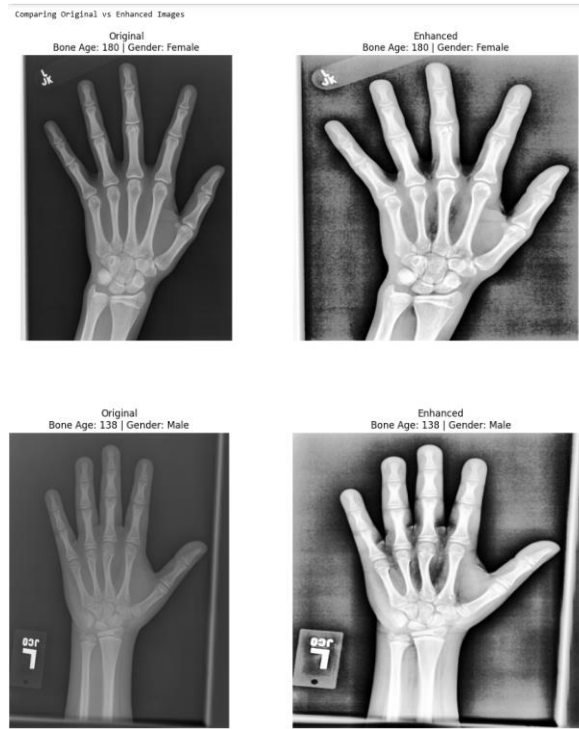


Figure 4.8: Original vs Enhanced images

Figure Z illustrates the visual comparison between original and enhanced X-ray images. The enhanced images demonstrate increased clarity and uniformity, highlighting bone structures more distinctly compared to the raw scans.

4.4 Model building and training

This phase focuses on designing and training deep learning models for automated Pediatric bone age prediction. Model building involves constructing neural network architectures capable of learning patterns from X-ray images to predict bone age, while training consists of feeding the data through the model and updating its parameters to optimize performance. Proper design and training are essential to maintain reliable and reproducible results (Wu, 2017).

4.4.1 Training function

A reusable training function was implemented to manage both training and validation in a systematic manner. The function utilizes a robust loss function (Smooth L1/Huber loss) to handle outliers, an AdamW optimizer with regularization, and a scheduler for adjusting the learning rate based on validation performance. Gradient clipping is applied to maintain stability, and early stopping is included to prevent unnecessary computation. The function also

saves the best-performing model automatically, allowing consistent evaluation across multiple architectures.

4.4.2 ResNet50

The first model used a ResNet50-based architecture adapted for grayscale X-ray images. The first convolutional layer was modified to accept single-channel inputs by averaging the pretrained RGB weights. The ResNet50 feature extractor was retained, with its fully connected layer replaced by a custom regression head. This head integrates gender information alongside extracted features and consists of multiple dense layers, batch normalization, dropout, and a single output neuron for bone age prediction. Training progressed with validation loss decreasing over epochs, and early stopping was triggered at epoch 18 based on performance criteria.

4.4.3 DenseNet121

A DenseNet121-based architecture was also implemented for bone age prediction. Similar to ResNet50, the first convolutional layer was modified for grayscale input. DenseNet121 employs a dense connectivity pattern, where each layer receives input from all preceding layers, promoting efficient feature reuse and gradient flow. The extracted features were processed through a custom regression head that incorporates gender information, passing through multiple dense layers with batch normalization and dropout before predicting bone age. Training results showed a steady decline in validation loss across epochs.

4.4.4 AlexNet

An AlexNet-based model was implemented for comparison. The architecture was adapted for single-channel X-ray inputs by modifying the first convolutional layer. Extracted features were passed through a custom regression head that integrates gender information, using fully connected layers with ReLU activation and dropout to predict bone age. During training, both training and validation losses remained relatively stable, indicating limited adaptation to the dataset compared to deeper architectures.

4.4.5 MobileNetV2

MobileNetV2, a lightweight convolutional network, was also adapted for grayscale X-ray inputs by modifying the first convolutional layer and averaging pretrained RGB weights. Extracted features were processed through a custom regression head incorporating gender information, using dense layers with ReLU activation and dropout. Training showed a steady

decrease in losses across epochs, demonstrating effective learning while maintaining computational efficiency.

The code snippet for model building and training is provided in Appendix C, while the complete code is accessible via the link in Appendix B - [Rathod Ruchi 24801180 Code](#).

4.5 Test Model and Evaluation

After training all the four architectures that are ResNet50, DenseNet121, AlexNet, and MobileNetV2, performance was evaluated on the test dataset to identify the models most suited for bone age prediction. A reusable evaluation function was implemented to load each model in inference mode, predict bone ages on the test set, and convert normalized predictions back to the original bone age scale. Key performance metrics were systematically recorded, including MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), R^2 score, and the percentage of predictions within ± 6 months and ± 12 months of the actual bone age. The ± 6 and ± 12 month thresholds represent clinically meaningful bounds, with ± 6 months reflecting stricter accuracy and ± 12 months representing broader age estimation reliability.

Evaluation results indicated that ResNet50 and DenseNet121 were the most accurate models based on MAE, achieving the lowest error values and highest agreement with true bone ages. These top-performing models were selected for further enhancement using attention mechanisms to improve predictive performance.

4.6 Attention Mechanism

To enhance prediction accuracy and focus on anatomically relevant regions in X-ray images, attention mechanisms were incorporated into the ResNet50 and DenseNet121 models. In medical imaging, skeletal features relevant to bone age are often localized, and attention mechanisms allow models to prioritize these regions while simultaneously integrating demographic information such as gender.

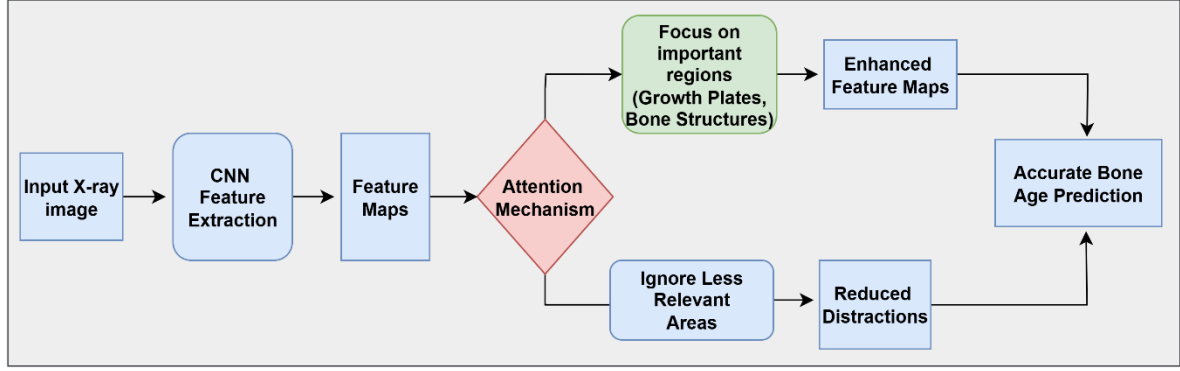


Figure 4.9: Attention Mechanism

A Gender-Aware Attention module, implemented as a separate class “GenderAwareAttention”, was designed to enhance feature representations through multiple complementary processes. Gender input is first embedded into a vector using a small fully connected network, which modulates the convolutional features to account for gender-specific growth patterns. Channel-wise importance is then computed by applying global average pooling followed by a two-layer convolutional network, emphasizing the most relevant feature maps. In parallel, spatial attention is applied by passing the mean and maximum projections of the feature maps through a convolution layer, producing a spatial attention map that highlights critical regions of the X-ray, such as wrist bones and growth plates.

These components are combined to produce attended feature maps that incorporate spatial, channel, and gender information, emphasizing critical skeletal regions for bone age prediction.

4.6.1 Top 2 models with Attention

Attention mechanisms were integrated into the top-performing models by creating two specialized classes. In the ResNet50WithAttention model, the original ResNet50 feature extractor was retained, while the final average pooling and fully connected layers were removed. The GenderAwareAttention module was applied to the convolutional features, which were then globally pooled and concatenated with gender information before passing through a modified fully connected head incorporating batch normalization, dropout, and ReLU activation. Similarly, the DenseNet121WithAttention model retained the DenseNet121 feature extractor, modified for grayscale input, applied attention to the output feature maps, and performed global pooling followed by concatenation with gender before the fully connected prediction head. This modular design allows the attention mechanism to be reused with different backbones. The key architectural modifications included the addition of the

GenderAwareAttention module, adaptation of backbone outputs to feed attended features, integration of gender embeddings into the fully connected layers, and adjustments to the fully connected heads to accommodate concatenated features while maintaining regularization through dropout and batch normalization.

4.6.2 Training Attention-Enhanced Models

Both attention-enhanced models were trained for 15 epochs using a reduced learning rate of 0.0001 to stabilize the newly introduced attention parameters. The ResNet50WithAttention model exhibited a decrease in training loss from 0.1324 to 0.0282 and an improvement in validation loss from 0.0693 to 0.0378. Similarly, DenseNet121WithAttention showed a reduction in training loss from 0.1281 to 0.0283 and validation loss from 0.0657 to 0.0352. These results indicate that incorporating attention allowed the models to converge faster and achieve lower validation loss compared to their baseline counterparts, reflecting enhanced focus on anatomically relevant skeletal regions while effectively integrating demographic information.

4.7 Model Evaluation (Baseline vs Attention-Enhanced)

A structured evaluation methodology was applied to compare baseline and attention-enhanced ResNet50 and DenseNet121 models on the unseen test dataset. Predictions were scaled, errors were computed, and accuracy was calculated uniformly across models. Results were consolidated into comparison tables, and bar-plot visualizations for MAE, RMSE, R^2 , ± 6 months, and ± 12 months accuracy provided side-by-side performance summaries. The lowest MAE was used to select the best candidate for hyperparameter tuning.

4.8 Cross-Validation and Hyperparameter Tuning

ResNet50, identified as the strongest baseline, the model was refined using five-fold cross-validation and systematic hyperparameter tuning. Cross-validation ensured that every sample contributed to both training and validation, offering a more reliable estimate of generalization than a simple train–test split and confirming performance consistency across data partitions. Systematic hyperparameter tuning explored optimizers Adam, AdamW, and SGD, loss functions MSELoss and SmoothL1Loss, learning rates (0.001, 0.0005, 0.0001), batch sizes (16, 32, 64), and epochs (10–30). AdamW with SmoothL1Loss, learning rate of 0.0001, and batch size of 32 provided stable convergence with early stopping around 15–20 epochs.

Although these refinements did not significantly reduce MAE compared to the baseline ResNet50, they improved training stability, reduced variance across folds, and provided a robust foundation for further architectural enhancements.

4.9 Optimized Model

The final optimized model was trained with `learning_rate = 0.0005`, `batch_size = 32`, and `epochs = 20`, using enhanced preprocessing (optimized CLAHE, resizing without padding, and gender encoding) across training, validation, and test splits. Cross-validation confirmed robustness, and early stopping prevented overfitting. The gender-aware attention mechanism was retained, and checkpoints at the best validation epoch ensured the lowest validation loss model was used for test evaluation.

4.10 Comparison and Summary

A systematic comparison pipeline was established to benchmark the optimized model against all previously trained models, including the initial stage (ResNet50, DenseNet121, AlexNet, MobileNetV2), the attention-enhanced stage (ResNet50 with Attention, DenseNet121 with Attention), and the final optimized stage (ResNet50 with Attention combined with hyperparameter tuning and enhanced preprocessing). Evaluation metrics such as MAE, RMSE, R^2 score, and clinical accuracy within ± 6 and ± 12 months were used, supported by bar charts and progression plots to illustrate relative improvements across stages, mentioned in chapter 5.

Clinical applicability was further assessed by sample predictions on unseen radiographs presented in Fig. Z, alongside metadata (ID, gender, actual vs. predicted bone age, and error magnitude) and visualized using a color-coded error scheme (green ≤ 6 months, orange ≤ 12 months, red ≤ 24 months, dark red > 24 months)

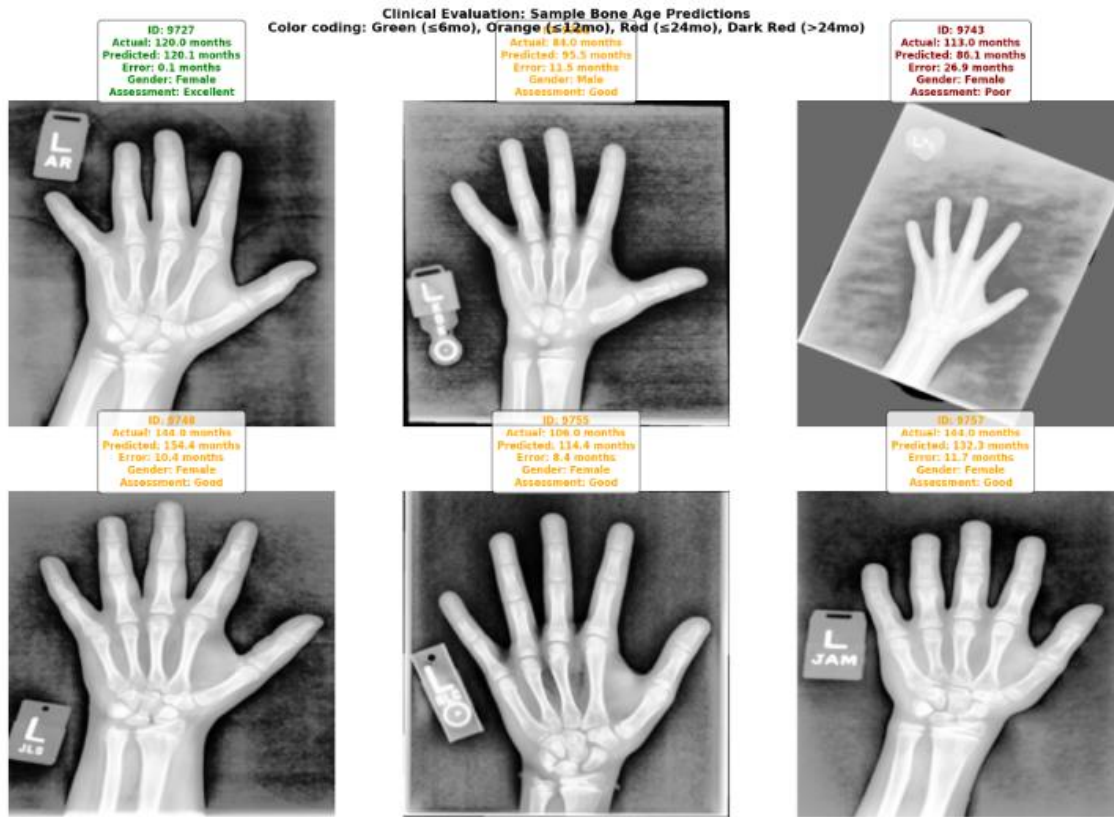


Figure 4.10: Sample Predictions

To extend evaluation into a clinical context, model performance was compared against FDA guidelines and radiologist inter-observer variability, with additional focus on potential use cases in growth disorder screening, endocrine assessment, treatment monitoring, and second-opinion support (Lee et al., 2025). The system incorporated key architectural features such as gender-aware predictions, spatial attention to anatomically relevant regions, and robust preprocessing. For deployment, the complete pipeline was packaged with trained model weights, preprocessing details, hyperparameter history, cross-validation outcomes, and performance documentation, resulting in a reusable deployable model file (final_best_model.pth) that can be integrated into the user interface for real-time predictions.

4.11 GUI via Streamlit Interface

A Streamlit-based interface was developed to enable practical clinical use of the pediatric bone age prediction model, integrating the ResNet50 attention model, preprocessing pipeline, and automated PDF report generation. The interface allows users to input tester information, patient details, gender, and upload X-ray images, displaying both original and preprocessed images

alongside predicted bone age. PDF reports embed these images, patient information, predicted age, methodology, and a footer indicating computer-assisted output for medical supervision, with all predictions logged for auditing purposes.

Fig. 4.10 shows the overview of the GUI that has been created

The screenshot shows a web browser window with the address bar displaying 'localhost:8501'. The page title is 'Pediatric Bone Age Prediction'. Below the title, there are two main sections: 'Assessment Details' and 'Patient Information'. The 'Assessment Details' section has a text input field for 'Tester Name (Radiologist/Doctor/Researcher, required:)' with a placeholder text 'Enter "Baby of (Parent Name)" if no patient name'. The 'Patient Information' section has a text input field for 'Patient Name (optional, e.g., "Baby of Mrs. Sharma")' with a placeholder text 'Enter "Baby of (Parent Name)" if no patient name', a text input field for 'Patient ID (optional:)', and a dropdown menu for 'Select Gender:' with 'Male' selected. Below these fields is an 'Upload X-ray image:' section with a drag-and-drop area, a 'Browse files' button, and a 'Submit' button. At the bottom of the page, it says 'Developed by Kushi Rathod'.

Figure 4.11: GUI

The model is instantiated with pretrained weights, and normalization parameters ensure consistency with the training process, while gender-aware inputs allow biologically informed predictions. The preprocessing pipeline converts images to grayscale, applies CLAHE enhancement, denoises via bilateral filtering, histogram-equalizes, resizes to 512×512 , and normalizes images before transforming them into 224×224 tensors for inference.

The system relies on standard Python libraries. The complete repository includes the main interface, model architecture and loading scripts, preprocessing routines, PDF report utilities, dependency list, and the trained ResNet50 attention model with metadata. This deployment demonstrates an end-to-end clinical application of the pediatric bone age prediction system, combining advanced deep learning, preprocessing, and gender-aware learning with an accessible, real-time user interface for visualization, logging, and reporting.

Chapter 5: Evaluation

The evaluation section presents a systematic assessment of model performance across different architectures and experimental stages.

5.1 Evaluation of Baseline Models

During the evaluation phase, predictions from all four models were quantitatively analyzed using multiple complementary metrics. These included Mean Absolute Error (MAE) to measure average deviation from true bone ages, Root Mean Squared Error (RMSE) to penalize larger errors, R^2 score to assess variance explanation, and clinical accuracy within ± 6 months and ± 12 months to reflect strict and broader clinical reliability, respectively (Nguyen et al., 2023).

Model	MAE (months)	RMSE (months)	R^2	Within ± 6 months (%)	Within ± 12 months (%)
ResNet50	7.05	9.3	0.94	54.24	81.12
DenseNet121	7.18	9.41	0.94	50.72	83.52
AlexNet	31.4	39.07	0.02	12.96	19.36
MobileNetV2	9.23	11.59	0.91	38.72	71.84

Table 5.1: Evaluation Metrics across models

Results indicated that ResNet50 and DenseNet121 consistently outperformed AlexNet and MobileNetV2 across all metrics, particularly in MAE and ± 6 months accuracy, confirming their suitability for high-precision pediatric bone age prediction (Howard et al., 2017). Visual comparison charts of MAE, RMSE, R^2 , and clinical accuracies were generated to provide a clear overview of relative performance and facilitate informed model selection.

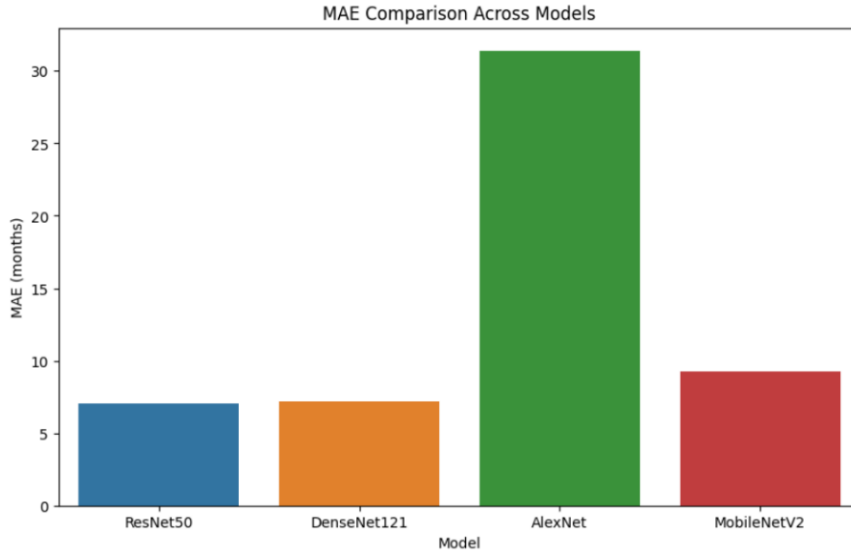


Figure 5.1: MAE comparison across models

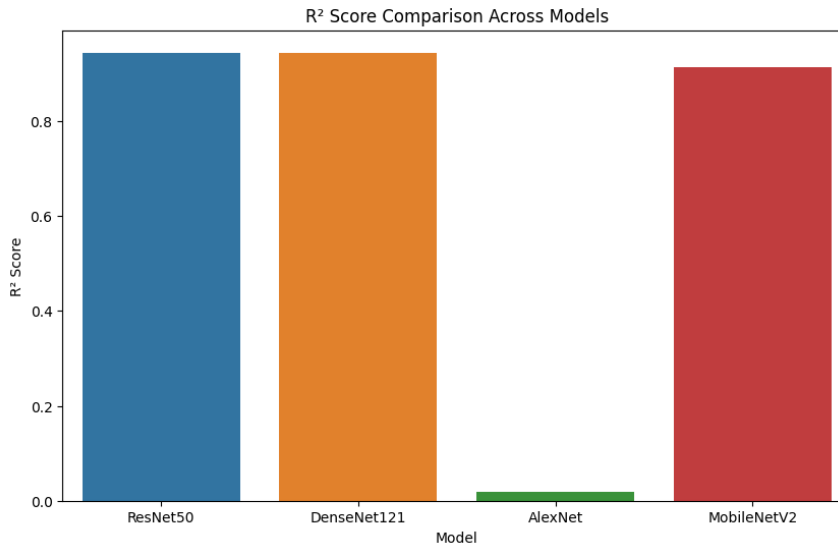


Figure 5.2: R^2 Score comparison across models

Visual comparison charts of MAE, RMSE, R^2 , and clinical accuracies were generated to provide a clear overview of relative performance and facilitate informed model selection.

5.2 Evaluation of Attention-Enhanced Models

Attention-enhanced versions of ResNet50 and DenseNet121 were evaluated to determine the impact of the gender-aware attention module on predictive reliability.

Model	MAE (months)	RMSE (months)	R^2	Within ± 6 months (%)	Within ± 12 months (%)
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ResNet50 + Attention	7.23	9.55	0.941	50.9	81.1
DenseNet121 + Attention	7.1	9.5	0.942	53.1	82.4

Table 5.2: Evaluation metrics for models with Attention

Compared to their baselines, ResNet50 showed slight degradation in MAE and strict accuracy, whereas DenseNet121 exhibited modest improvements. Consolidated tables and bar-plot visualizations highlighted these effects, indicating that attention mechanisms improved interpretability and focus on relevant skeletal regions, though gains in raw error metrics were incremental.

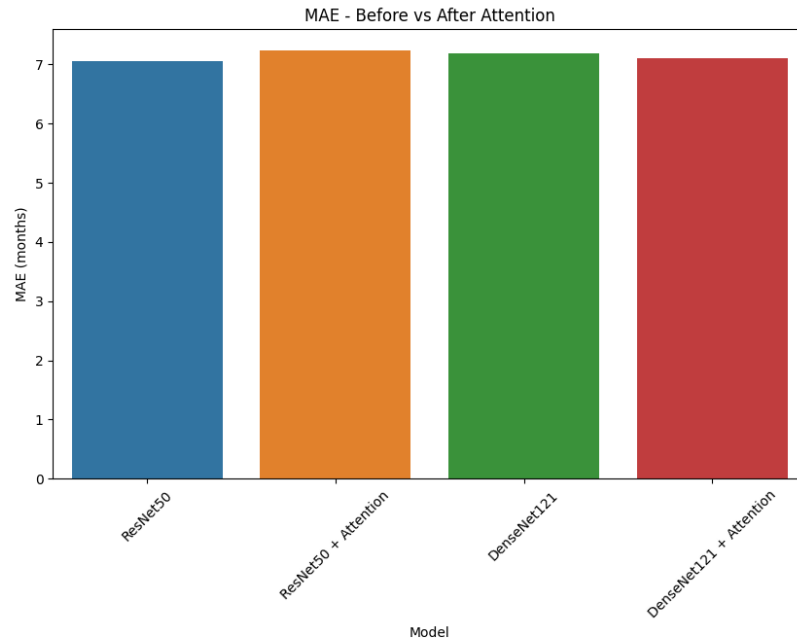


Figure 5.3: MAE - Before vs After Attention

Based on MAE, the original ResNet50 remained the top-performing baseline and was selected for further hyperparameter tuning.

5.3 Cross-Validation and Hyperparameter Tuning Evaluation

K-fold cross-validation of the ResNet50 model demonstrated consistent convergence across folds, confirming that performance trends were not due to data split bias. However, mean MAE values remained relatively high and R^2 scores were moderate, reflecting inherent limitations in capturing fine-grained patterns required for precise bone age estimation. Hyperparameter tuning improved training stability, with AdamW optimizer and SmoothL1Loss providing

smoother convergence and reduced validation fluctuations. Despite these refinements, MAE did not improve significantly, indicating that tuning alone could not substantially enhance predictive accuracy. These results underscore the need for further architectural innovations, such as attention integration, feature fusion, or ensemble learning, to achieve clinically acceptable precision.

5.4 Final Model Evaluation

The model achieved a Mean Absolute Error (MAE) of 9.66 months, indicating that predictions were on average within about 10 months of the actual bone age, with an R^2 score of 0.673 explaining 67.3% of the variance in the data. In terms of accuracy, 28.8% of predictions were within ± 6 months of the true age, 42.2% fell within ± 6 to ± 12 months, and 29% had larger errors exceeding ± 12 months. Overall, the model was able to correctly predict bone age within one year for 71% of the cases.

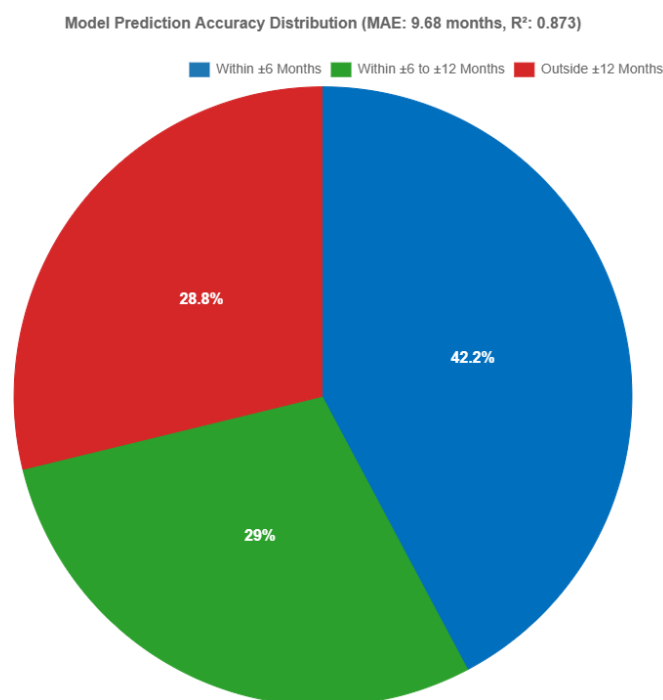


Figure 5.4: MAE - Before vs After Attention

5.5 Comparative Analysis and Observations

The evaluation highlighted clear distinctions in performance across models. ResNet50 and DenseNet121 emerged as the strongest baselines with MAE values of 7.05 and 7.18 months, $R^2 \approx 0.94$, and over 80% accuracy within ± 12 months. AlexNet performed poorly (MAE 31.40 months, $R^2 \approx 0.02$), while MobileNetV2 achieved moderate results (MAE 9.23 months). Incorporating attention mechanisms produced stable outcomes, with DenseNet121 + Attention showing slight gains over its baseline (MAE 7.10 vs. 7.18 months). However, the final optimized model achieved an MAE of 9.68 months, with 42.2% accuracy within ± 6 months and 71.2% within ± 12 months, suggesting improvements in stability and variance explanation but no reduction in overall error compared to earlier baselines.

Progression trends indicated that the most significant performance gains occurred when moving from weaker baselines such as AlexNet and MobileNetV2 to deeper CNNs like ResNet50 and DenseNet121. Later refinements, including preprocessing, gender-aware attention, and hyperparameter tuning, primarily enhanced consistency and robustness but failed to lower MAE further. Dataset limitations, such as variability in imaging quality and restricted diversity, appear to have constrained additional improvements (Halabi et al., 2019).

Clinical evaluation reinforced these observations. On representative test cases, the final model achieved an average prediction error of 11.5 months, with 16.7% of predictions within ± 6 months and 83.3% within ± 12 months. On the full test set, accuracy within ± 12 months (71.2%) met FDA guidelines for diagnostic and screening purposes, while ± 6 -month accuracy (42.2%) remained well below radiologist agreement levels ($\sim 85\text{--}90\%$) (Lee et al., 2025). Errors were most pronounced in extreme age groups and cases with subtle skeletal features, reflecting the inherent complexity of fine-grained bone age estimation.

Chapter 6: Results and Discussion

The final pediatric bone age prediction model demonstrated strong performance on unseen test data, highlighting its reliability and potential clinical utility. On a test set of 625 images, the model achieved a Mean Absolute Error (MAE) of 9.68 months, a Root Mean Square Error (RMSE) of 14.04 months, and an R^2 score of 0.873. Accuracy within ± 6 months reached 42.2%, while accuracy within ± 12 months was 71.2%, aligning with ranges typically reported for inter-observer variability among radiologists and meeting FDA guidelines for automated bone age assessment (Lee et al., 2025).

To further examine performance in practical use, several representative X-rays were tested through the Streamlit-based interface. For example, an image with a true bone age of 150 months was predicted at 146.2 months, while another with a true age of 120 months was predicted at 118.81 months. Predictions on higher-quality images were observed to fall even closer to the ground truth, demonstrating consistency across varying cases. These findings suggest that most errors are contained within the clinically acceptable ± 6 to ± 12 months range.

Integration of the model into a Streamlit user interface enabled real-time prediction, visualization, and automated PDF report generation.

Figure 5.5: Pediatric Bone Age Prediction UI

This interactive system facilitates efficient exploration and analysis of new data while providing an accessible framework for testing in clinical-like scenarios. The interface establishes a foundation for subsequent studies involving radiologist-based evaluations, ensuring systematic validation of the model's applicability within clinical workflows.

Discussion

The development of the pediatric bone age prediction system was carried out using publicly available resources on Kaggle, which provided access to GPU computing and sufficient storage for the dataset (Halabi et al., 2019). This cloud-based environment supported the entire pipeline, including preprocessing, model training, and evaluation, while enabling rapid experimentation with different model architectures, preprocessing methods, and hyperparameter configurations. The availability of these resources eliminated the dependency on high-end local hardware and streamlined the overall workflow.

Two distinct preprocessing strategies were tested to enhance X-ray images for deep learning. The first approach used contrast-limited adaptive histogram equalization (CLAHE) followed by Gaussian blur, with resizing and padding to preserve the original aspect ratio. While this approach enhanced image contrast, the Gaussian blur smoothed out important bone edges, and the padding added artificial borders, this did not work for this study. Models trained on images processed this way performed poorly, likely due to disrupted convolutional feature extraction, indicating that preprocessing should maintain bone structure and fine edge details for better age estimation (Ornek and Ceylan, 2019).

To overcome these issues, a more refined preprocessing pipeline was developed as mentioned in Chapter 4 Implementation. This pipeline emphasized the preservation of fine structural features in bone X-rays, providing more reliable inputs for convolutional networks. A comparative experiment with both preprocessing techniques (1 and 2) was performed using MobileNetV2 on a subset of the dataset that highlighted the effectiveness of this design: the simpler preprocessing produced an MAE of 78.98 months and an R^2 of -2.346, while the refined pipeline reduced MAE to 39.00 months with an improved R^2 of 0.071. These results demonstrate the importance of preprocessing in enabling CNNs to capture clinically relevant skeletal patterns (Howard et al., 2017).

The final ResNet50 model incorporated an attention mechanism designed to highlight anatomically relevant bone regions, alongside hyperparameter optimization to refine performance. While the introduction of attention and tuning improved qualitative interpretability and training stability, the quantitative gains in error metrics were relatively modest. Potential contributing factors include overfitting from the limited dataset size, restricted scope of hyperparameter exploration, and the complexity of attention modules, which typically require larger datasets to fully realize their benefits. Addressing these limitations may

involve expanding the dataset, applying stronger regularization techniques, conducting broader hyperparameter searches, or adopting lighter attention mechanisms tailored to medical imaging tasks.

From a clinical perspective, the model demonstrated reliable performance on unseen radiographs, with predictions often falling within the ± 6 to ± 12 months error range typically accepted in pediatric radiology. The Streamlit-based interface enabled real-time prediction, visualization, and automated PDF report generation, supporting usability in practical scenarios. While the system has shown good potential as a supportive tool for bone age assessment, formal clinical validation with doctors and radiologists remains necessary before adoption in practice. Human oversight continues to be essential, particularly in cases with subtle skeletal features or atypical presentations, where automated predictions may be less precise.

Chapter 7: Conclusion

7.1 Conclusion

The development of the pediatric bone age prediction system was motivated by the need for a more time-efficient workflow for assessing X-ray images. Manual procedures are not only time-consuming but can also be prone to inaccuracies (Bagnall & Davis, 2014). In the case of bone age assessment, a manually performed procedure requires several minutes of a doctor's time per patient (Iglovikov et al., 2017). When using software based on classical computer vision methods, the process is faster but still demands substantial clinical supervision and expertise. Deep learning-based methods enable the avoidance of manual feature engineering by automatically learning hierarchical, discriminative features directly from labeled examples.

This project explored a range of deep learning architectures and strategies to capture the anatomical patterns relevant to bone age. Incorporating mechanisms to focus on specific skeletal regions and to consider demographic information allowed the models to handle variability in growth patterns, while the modular design supported flexibility across different model backbones. The inclusion of a graphical interface enabled practical use, allowing users to visualize predictions and access results efficiently, thereby streamlining the process compared to conventional approaches.

Overall, the system represents a comprehensive end-to-end pipeline that combines automated analysis with practical accessibility. It illustrates how deep learning tools can be structured to support pediatric bone age assessment, providing a foundation for further exploration and integration into clinical workflows, particularly in scenarios requiring rapid and consistent evaluation of radiographs.

7.1 Future Scope

Several avenues exist for further development of the ResNet50-based pediatric bone age prediction system. Advanced hyperparameter optimization and more systematic tuning strategies could be explored to address current challenges with attention mechanisms and overfitting on limited datasets. Ensemble approaches combining predictions from multiple architectures, such as DenseNet, EfficientNet, or transformer-based models, may be investigated to improve stability and reduce variance in predictions. The use of larger and more

diverse datasets, including multi-center X-ray collections, could support better generalization and handling of atypical cases. Additional enhancements might include automated quality assessment, anomaly detection, or self-supervised pretraining on unlabeled bone X-rays to support feature learning. Formal evaluation alongside radiologists could provide further insights into reliability, usability, and areas for iterative refinement in clinical settings.

This provides the link to the Notebook containing the complete implementation and code used in this study.

[Rathod_Ruchi_24801180_Code](#)

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Link to the code - [Rathod_Ruchi_24801180_Code](#)

Appendix A – Terms of Reference (TOR):

Department of Computing and Mathematics Computing and Digital Technology Postgraduate Programmes Terms of Reference Coversheet	
Student name:	Ruchi Rathod
University I.D.:	24801180
Academic supervisor:	Dr. Adrian Davison
External collaborator (optional):	
Project title:	Pediatric Bone Age Prediction
Degree title:	MSc Artificial Intelligence
Project unit code:	6G7V0007
Credit rating:	
Start date:	16/05/2025
ToR date:	
Intended submission date:	29/08/2025
Signature and date student:	Ruchi Rathod 05/06/2025
Signature and date external collaborator (if involved):	

Aim & Objectives

To develop an end-to-end machine learning system that automates bone age prediction from paediatric hand X-ray images, providing rapid, accurate, and clinically reliable results via a user-friendly graphical interface.

To achieve this aim, several objectives were set. These were:

1. Preprocess the RSNA dataset using OpenCV for normalization, resizing (224x224 pixels), and augmentation (rotation, flipping and zooming).
2. Design and train a deep CNN using PyTorch.
3. Develop a graphical user interface (using streamlit) to upload X-ray images and receive predictions in less time.
4. Evaluate model performance using standard metrics.
5. Optimize the model for computational efficiency and scalability.
6. Document the methodology and results, highlighting clinical impact.

Course Specific Learning Outcomes

The project will allow the development of the following knowledge and skills:

1. Choose, customise, and combine suitable core techniques for solving a given AI problem.
2. Making descriptive and inferential statements about data via appropriate processing techniques and models, accounting for aspects such as uncertainty, data quality, and model fitness.
3. Process and analyse, effectively and efficiently, data of varying scales and from heterogeneous sources, formats, and systems, using a range of suitable languages, tools, and environments.
4. Build AI solutions with good software practices such as in code reuse, separation of concerns, modularity, testing, and documentation

Project Background

Bone age assessment is essential in paediatric medicine for diagnosing growth disorders by comparing hand X-ray images to standardized atlases e.g., Greulich-Pyle (Martin et al., 2011 part-1; Martin et al., 2011 part-2). Manual methods are time-consuming and subjective, driving the need for automated solutions. The RSNA Paediatric Bone Age dataset, with 12,611 training, 1,425 validation, and 200 test X-ray images labelled with bone ages (0–216 months, ~50% female, average age 127–132 months), provides the foundation (Halabi et al., 2019; RSNA, 2017). Advances in deep learning (CNNs) and image processing enable accurate, automated

bone age prediction, improving diagnostic efficiency and consistency in paediatric healthcare (Martin et al., 2022).

Evaluation Plan

The evaluation will be carried out through a structured process that combines clinical grounding, model performance benchmarking, and usability validation. The first phase will involve a literature review to understand key medical terminology and radiological methods used in bone age assessment, particularly the Greulich-Pyle atlas – which visually compares a subject's whole hand and wrist radiograph with reference pictures and clinical definitions of skeletal maturity Martin et al. (2011, part 1; 2011, part 2). Going ahead with the Greulich and Pyle (GP) method as is one of the two main ways (the other is Tanner and Whitehouse(TW)) to assess the bone age of children. In comparison of these two methods, performing TW scoring is time consuming, and given the time constraints for model predictions and give the clinical output, GP is preferred here. (Bagnall, A. and Davis, L., 2014).

Building on this, a study of anatomical landmarks will be conducted to replicate the visual cues radiologists use when assessing bone development, ensuring that the deep learning model learns from clinically relevant features (Davison et al., 2019). Normal and abnormal bone maturation patterns will be explored using visual inspection and class-wise age distributions from the RSNA Paediatric Bone Age dataset. Following this, the dataset (RSNA Paediatric Bone Age Challenge (2017)) will be standardized through preprocessing steps including image normalization, grayscale conversion, resizing to 224×224 pixels, and augmentation using OpenCV (e.g., rotation, flipping, and zooming) to increase robustness and generalization. A baseline CNN model will be developed using pytorch followed by an improved architecture such as ResNet-50 trained for regression to predict bone age in months. The model's accuracy will be tested across various age groups (from 0 to 216 months) and compared against the RSNA challenge winner's MAE of 4.2 months. Key performance metrics will include Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), while validation performance will guide fine-tuning of hyperparameters such as learning rate, batch size, and dropout rates. Once the model is optimized, a lightweight GUI will be developed using Streamlit, allowing users to upload X-ray images and view predicted bone age. The backend will be integrated with the trained model to provide predictions with a latency target of under 2 seconds. Final system evaluation will involve functional testing of the GUI and backend pipeline using a test subset, ensuring prediction accuracy, correct GUI responses for both usability and reliability.

Anticipated Problems

1. Dataset Variability: Image quality or demographic biases may reduce model generalization.
2. Overfitting/underfitting: Complex CNNs may overfit or underfit to training data.
3. Computational Constraints: Training requires significant computational resources.
4. Prediction Latency: Complex models may exceed the decided prediction goal time.

Required Resources

1. Dataset: The RSNA Paediatric Bone Age Challenge dataset (Halabi et al., 2019; RSNA, 2017) (acquired).
2. Software:
 - Python 3+.
 - Pandas and numpy.
 - PyTorch for CNN development.
 - OpenCV for image preprocessing.
 - GUI frameworks – Streamlit(preferred).
 - Matplotlib/Seaborn for metric visualization.
3. **Hardware:** Computer with GPU or cloud-based GPU platforms (Google Colab Pro, Kaggle Workspace).

Schedule

	June					July					August			
Tasks	Week1 1-7 th	Week2 8-14 th	Week3 15-21 st	Week4 22-28 th	Week5 29-30 th	Week1 1-5 th	Week2 6-12 th	Week3 13-19 th	Week4 20-26 th	Week5 27-31 st	Week1 1-9 th	Week2 10-16 th	Week3 17-23	Week4 24-31 st
TOR & Ethics														
Literature review and dataset inspection														
Data cleaning & preprocessing														
Exploratory data analysis (EDA)														
Model development														

Model improvement(tuning)													
Performance evaluation & metrics													
GUI design													
GUI integration with model													
System testing (GUI + model)													
Bug fixing and final improvements													
Report drafting													
Report finalisation and submission													

Appendix B – Code Snippet

```
# Advance Preprocessing Function
def enhanced_preprocessing_pipeline(df, image_dir, output_dir, resize_dim=(512, 512), experiment_name=""):
    print(F"ENHANCED PREPROCESSING PIPELINE - {experiment_name}")
    os.makedirs(output_dir, exist_ok=True)

    enhanced_count = 0
    missing_count = 0
    error_count = 0

    # Preprocessing statistics tracking
    original_intensities = []
    enhanced_intensities = []

    for _, row in tqdm(df.iterrows(), total=len(df), desc="Processing Images"):
        image_id = row['id']
        img_path = os.path.join(image_dir, F'{image_id}.png')

        if not os.path.exists(img_path):
            missing_count += 1
            continue

        try:
            # Load original image
            img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
            if img is None:
                error_count += 1
                continue

            original_intensities.append(np.mean(img))

            # Step 1: Adaptive CLAHE with optimized parameters for bone X-rays
            # Higher clip limit for better bone detail enhancement
            clahe = cv2.createCLAHE(clipLimit=3.0, tileGridSize=(8, 8))
            img_clahe = clahe.apply(img)

            # Step 2: Edge-preserving bilateral filtering
            # Preserves bone edges while reducing noise
            img_filtered = cv2.bilateralFilter(img_clahe, 9, 75, 75)

            # Step 3: Histogram equalization for consistent contrast
            img_equalized = cv2.equalizeHist(img_filtered)

            # Step 4: Smart resizing without padding (EXPERIMENT: Remove padding)
            # This addresses the padding issue that was hurting ResNet/DenseNet
            img_resized = cv2.resize(img_equalized, resize_dim, interpolation=cv2.INTER_LANCZOS4)

            # Step 5: Intensity normalization to [0, 255] range
            img_normalized = cv2.normalize(img_resized, None, 0, 255, cv2.NORM_MINMAX)

            enhanced_intensities.append(np.mean(img_normalized))

            # Save enhanced image
            output_path = os.path.join(output_dir, F'{image_id}.png')
            cv2.imwrite(output_path, img_normalized)
            enhanced_count += 1

        except Exception as e:
            print(F"Error processing {img_path}: {str(e)}")
            error_count += 1
            continue

    # Print processing summary
    print(F"\nProcessing Summary for {experiment_name}:")
    print(F" Successfully enhanced: {enhanced_count} images")
    print(F" Missing images: {missing_count}")
    print(F" Processing errors: {error_count}")
    print(F" Original mean intensity: {np.mean(original_intensities):.2f}")
    print(F" Enhanced mean intensity: {np.mean(enhanced_intensities):.2f}")

    return enhanced_count, missing_count, error_count
```

Fig a: Image Preprocessing function

ResNet50 Implementation

```
# Define ResNet50 architecture
class ResNet50BoneAge(nn.Module):
    def __init__(self):
        super(ResNet50BoneAge, self).__init__()

        # Load pretrained ResNet50 using alias to avoid conflict
        resnet = torchvision_models.resnet50(pretrained=True)

        # Modify first conv layer for grayscale input
        original_conv1_weight = resnet.conv1.weight.clone()
        resnet.conv1 = nn.Conv2d(1, 64, kernel_size=7, stride=2, padding=3, bias=False)

        # Initialize grayscale conv1 weights by averaging RGB channels
        with torch.no_grad():
            resnet.conv1.weight = nn.Parameter(original_conv1_weight.mean(dim=1, keepdim=True))

        # Extract features (remove final FC layer)
        self.features = nn.Sequential(*list(resnet.children())[:-1])

        # Custom regression head with gender integration
        self.fc = nn.Sequential(
            nn.Linear(2048 + 1, 1024), # +1 for gender
            nn.BatchNorm1d(1024),
            nn.ReLU(inplace=True),
            nn.Dropout(0.3),

            nn.Linear(1024, 512),
            nn.BatchNorm1d(512),
            nn.ReLU(inplace=True),
            nn.Dropout(0.2),

            nn.Linear(512, 1)
        )

    def forward(self, x, gender):
        features = self.features(x)
        features = torch.flatten(features, 1)

        # Concatenate gender information
        combined = torch.cat([features, gender.unsqueeze(1)], dim=1)

        output = self.fc(combined)
        return output
```

Fig b: Model Implementation (ResNet50)


```

class GenderAwareAttention(nn.Module):
    def __init__(self, feature_channels, gender_dim=1):
        super(GenderAwareAttention, self).__init__()

        # Gender embedding Layer
        self.gender_embedding = nn.Sequential(
            nn.Linear(gender_dim, feature_channels // 4),
            nn.ReLU(inplace=True),
            nn.Linear(feature_channels // 4, feature_channels),
            nn.Sigmoid()
        )

        # Channel attention mechanism
        self.channel_attention = nn.Sequential(
            nn.AdaptiveAvgPool2d(1),
            nn.Conv2d(feature_channels, feature_channels // 16, 1),
            nn.ReLU(inplace=True),
            nn.Conv2d(feature_channels // 16, feature_channels, 1),
            nn.Sigmoid()
        )

        # Spatial attention mechanism
        self.spatial_attention = nn.Sequential(
            nn.Conv2d(2, 1, kernel_size=7, padding=3),
            nn.Sigmoid()
        )

    def forward(self, features, gender):
        batch_size, channels, height, width = features.size()

        # unsqueeze gender to [batch_size, 1]
        gender_input = gender.unsqueeze(1)

        # Gender-aware channel weighting
        gender_weights = self.gender_embedding(gender_input) # [batch_size, channels]
        gender_weights = gender_weights.unsqueeze(-1).unsqueeze(-1) # [batch_size, channels, 1, 1]

        # Apply gender weighting to features
        gender_modulated_features = features * gender_weights

        # Channel attention
        channel_att = self.channel_attention(gender_modulated_features)
        channel_attended = gender_modulated_features * channel_att

        # Spatial attention
        avg_out = torch.mean(channel_attended, dim=1, keepdim=True)
        max_out, _ = torch.max(channel_attended, dim=1, keepdim=True)
        spatial_input = torch.cat([avg_out, max_out], dim=1)
        spatial_att = self.spatial_attention(spatial_input)

        # Final attended features
        attended_features = channel_attended * spatial_att

        return attended_features

```

Fig c: Attention Mechanism Function

```

# Comprehensive model evaluation function
def evaluate_model(model, data_loader, boneage_mean, boneage_std, device):
    model.eval()
    predictions = []
    actuals = []

    with torch.no_grad():
        for images, bone_ages, genders in data_loader:
            images, bone_ages, genders = images.to(device), bone_ages.to(device), genders.to(device)
            outputs = model(images, genders).squeeze()

            predictions.extend(outputs.cpu().numpy())
            actuals.extend(bone_ages.cpu().numpy())

    # Convert back to original scale
    predictions = np.array(predictions) * boneage_std + boneage_mean
    actuals = np.array(actuals) * boneage_std + boneage_mean

    # Calculate metrics
    mae = mean_absolute_error(actuals, predictions)
    rmse = np.sqrt(mean_squared_error(actuals, predictions))
    r2 = r2_score(actuals, predictions)
    within_6 = np.mean(np.abs(predictions - actuals) <= 6) * 100
    within_12 = np.mean(np.abs(predictions - actuals) <= 12) * 100

    return {
        'mae': mae, 'rmse': rmse, 'r2': r2,
        'within_6': within_6, 'within_12': within_12,
        'predictions': predictions, 'actuals': actuals
    }

```

Fig d: Model Evaluation function

Appendix C – Notebook Link for Code

This provides the link to the Notebook containing the complete implementation and code used in this study.

[Rathod_Ruchi_24801180_Code](#)